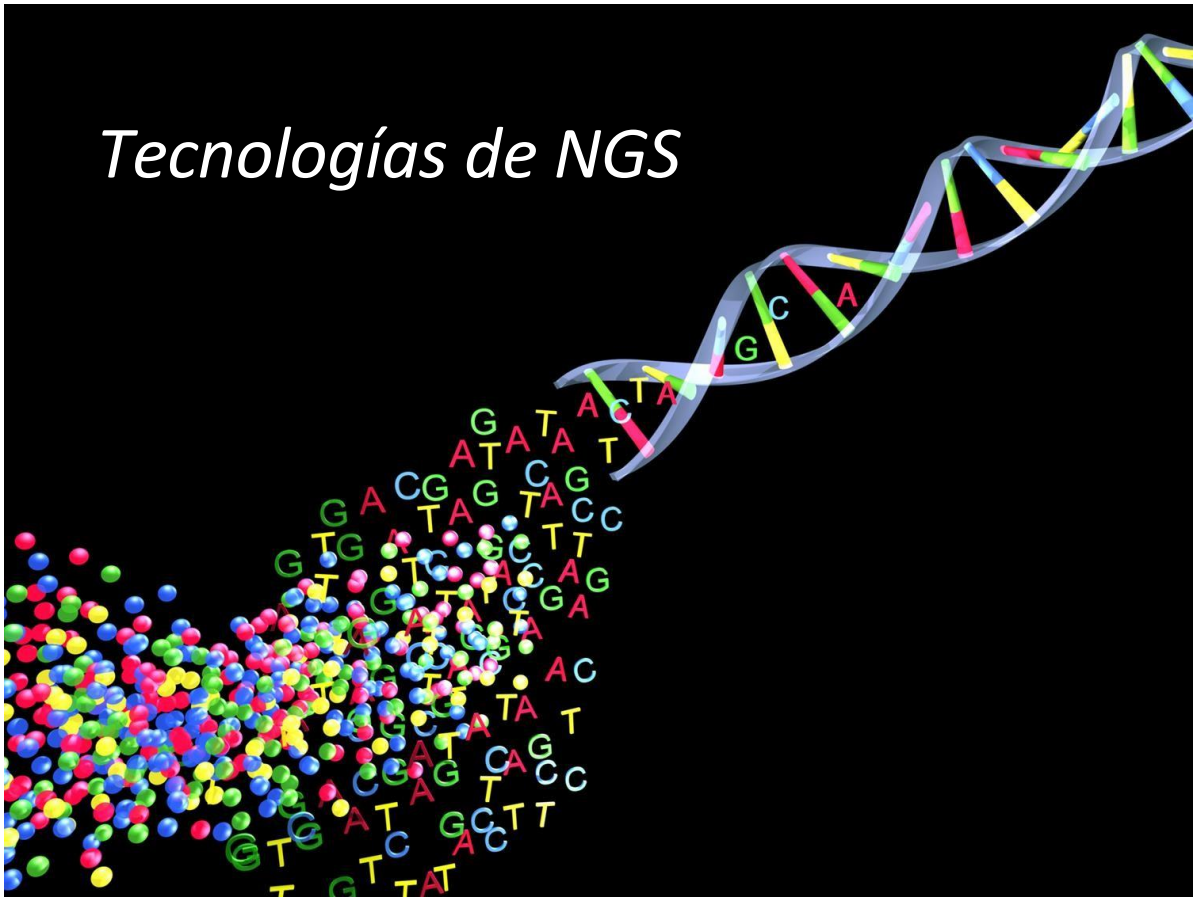


Begoña Aguado Orea
Centro de Biología Molecular Severo Ochoa
baguado@cbm.csic.es

El **Centro de Biología Molecular Severo Ochoa** (CBM) es un centro de investigación multidisciplinar en el área de la biología molecular. Centro mixto perteneciente al Consejo Superior de Investigaciones Científicas (CSIC) y a la Universidad Autónoma de Madrid (UAM).



NO es un centro de investigación especializado
Amplia diversidad de temas de investigación y organismos

Nuestra solución: Análisis personalizado

Servicio de Análisis Biocomputacional (SABio)

Análisis computacional
Soluciones personalizadas para datos multiómicos: genómica, transcriptómica, epigenómica, proteómica, metabolómica...

Asesoramiento científico y técnico
Desde el diseño experimental hasta la interpretación y el reporte de resultados.

Gestión y envío seguro de datos
Envío de datos a ENA/EGA y mantenimiento de la privacidad como intermediarios (*brokers*) de ENA (EMBL-EBI).

Formación personalizada
Cursos a la medida sobre análisis e interpretación de datos ómicos.

Desarrollo de software a medida
Pipelines y *scripts* personalizados para necesidades bioinformáticas complejas.



- **Cursos personalizados**

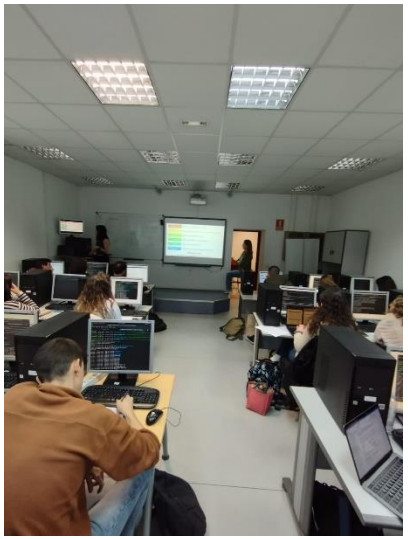
- PTI+Ciencia Digital → Videos ”Pildoras” análisis de datos de NGS: RNAseq & DNAseq

https://digital.csic.es/handle/10261/378652

-Gabinete de Formación CSIC:

- Análisis de datos de NGS

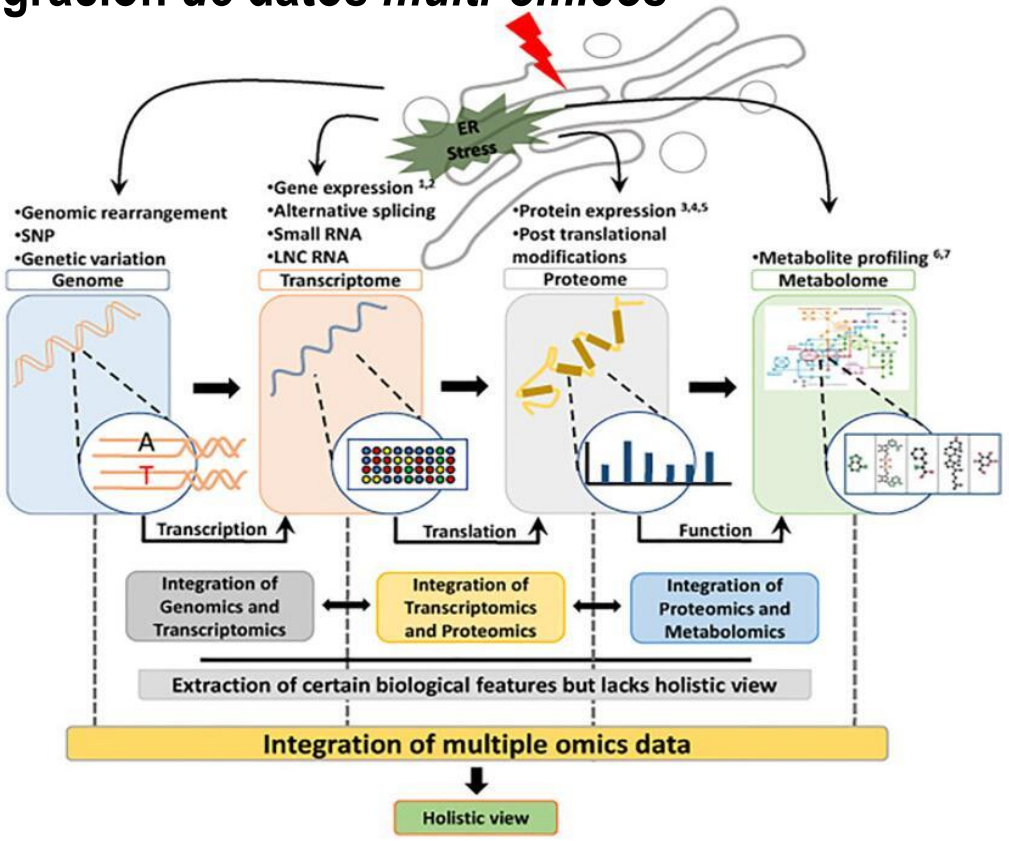
- Análisis de datos de scRNAseq



Integración *de* datos *multi-ómicos*

Tecnologías ómicas (sufijo deriva del griego *-oma* que indica *conjunto* o *masa*):

- **Reducción de Dimensiones (Estadística):** *Ejemplo:* **MOFA+** (Factorización de matrices / factores latentes).
- **Modelado basado en Redes (Grafos):** *Ejemplo:* **SNF** (Fusión de redes de similitud de pacientes).
- **Conocimiento Previo (Bases de datos):** *Ejemplo:* **Pathway Analysis** (Mapas de KEGG/Reactome).
- **Machine Learning (IA/Deep Learning):** *Ejemplo:* **Autoencoders multimodales** y Redes de Grafos (**GNN**).



https://www.sciencedirect.com/science/article/pii/S0167488922000015



https://pti-cienciadigital.csic.es/
pti-cienciadigital@csic.es

Ciencia y Tecnología de ácidos nucleicos: 70 años de Historia



1953 **DNA structure** (Watson y Crick)

1977: Sanger develops a sequencing method

1981: Moloney sarcoma virus (MSV) (5.8kb), Mitochondrial human genome (16.5kb)

1985: Mullis y col. describe the PCR

1987: Applied Biosystems Inc introduce to market the first authomatic sequencer based on fluorescent terminator dyes

1995: *Haemophilus influenzae* (1.8Mb) and *Mycoplasma genitalium* (0.58Mb)

1998: PE Biosystems Inc gets into the market the first capillary automatic sequencer

1998: *Caenorhabditis elegans* (97Mb), MHC class III human-mouse (0.9Mb)

2000: *Drosophila melanogaster* (120Mb) and *Arabidopsis thaliana* (125Mb)

2001: Draft of the human genome (3000Mb)

2003 **Complete sequence of a human genome: 3000 M bases**

3.000M euros 10 years

2004: Masive sequencing or second generation (Roche 454, Illumina, SOLID).

Masive sequencing or third generation (PacBio, Ion Proton, Oxford Nanopore, …)

2014: HiSeqX, Ion Proton: Human genome for 760€ (1000$) in half an hour or a day.

2015: Human population projects: UK, USA… 100.000s of human genomes.

2020: COVID-19

2023: Genomes, tissues,…single cell projects, ….

Ciencia y Tecnología de ácidos nucleicos: 70 años de Historia



1953 **DNA structure** (Watson y Crick)

Computing needs

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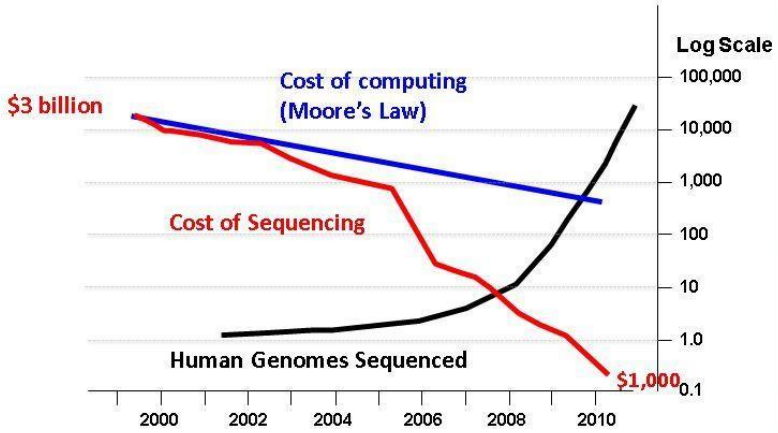
Computing complexity

2023: Genomes, tissues,…single cell projects, ….

Sequenced Genomes

Adapted from The Economist

The Sequencing Explosion



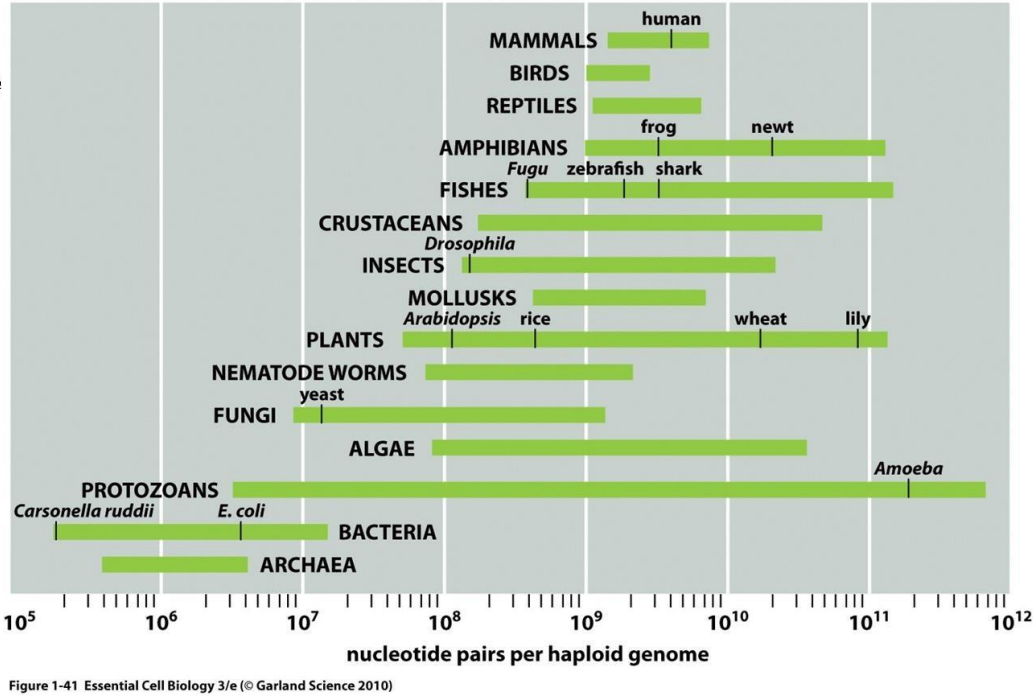
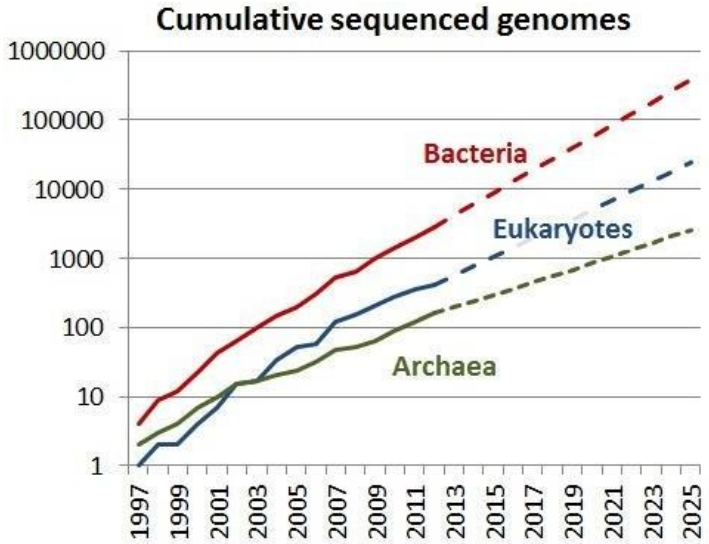


Figure 1-41 Essential Cell Biology 3/e (© Garland Science 2010)



Eukaryotes (30.530); Prokaryotes (567.228); Viruses (66.429); Plasmids (49.499); Organelles (3.020)

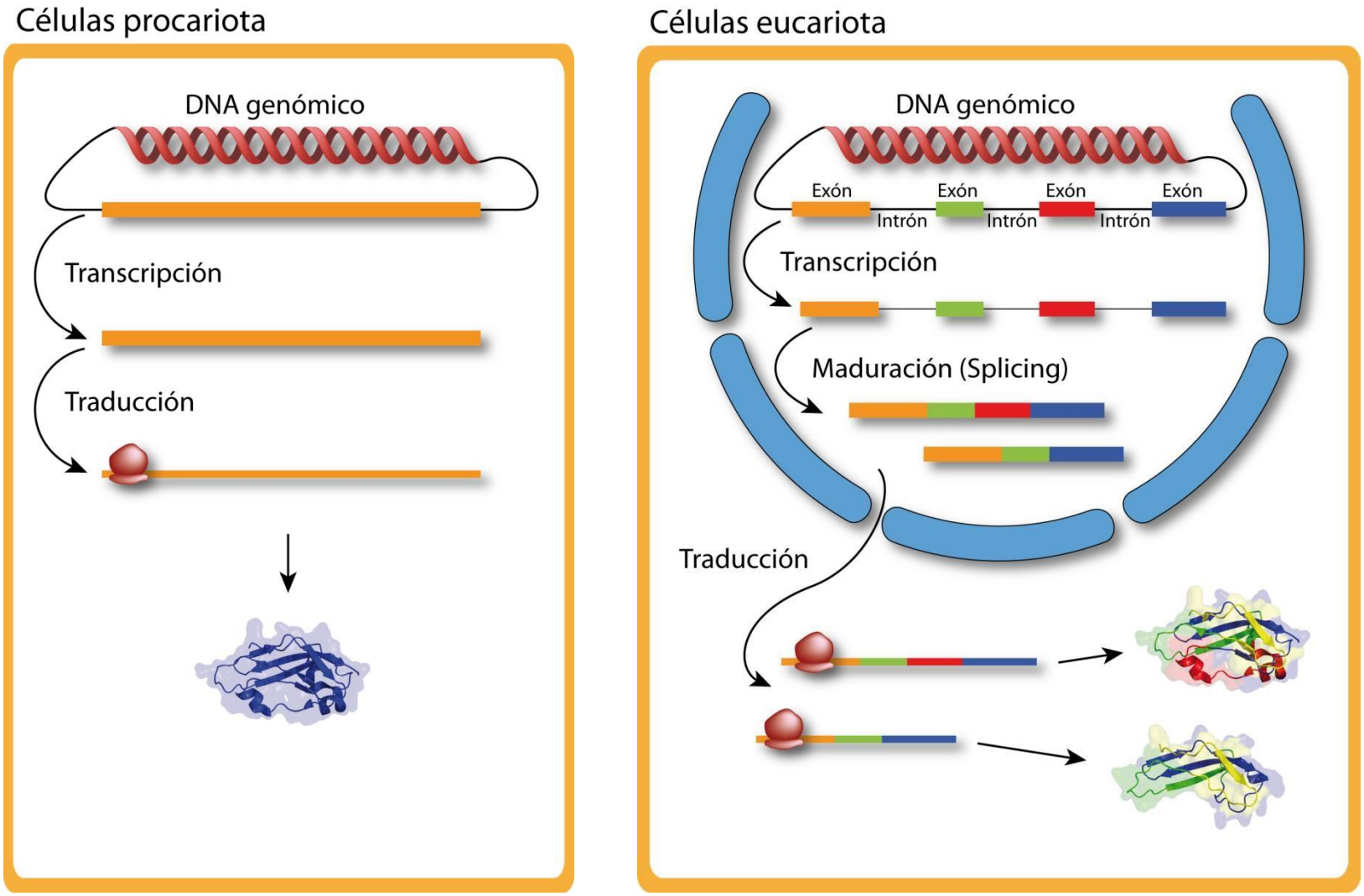
Human: 1.067 assemblies

Tamaño de genomas en diferentes organismos

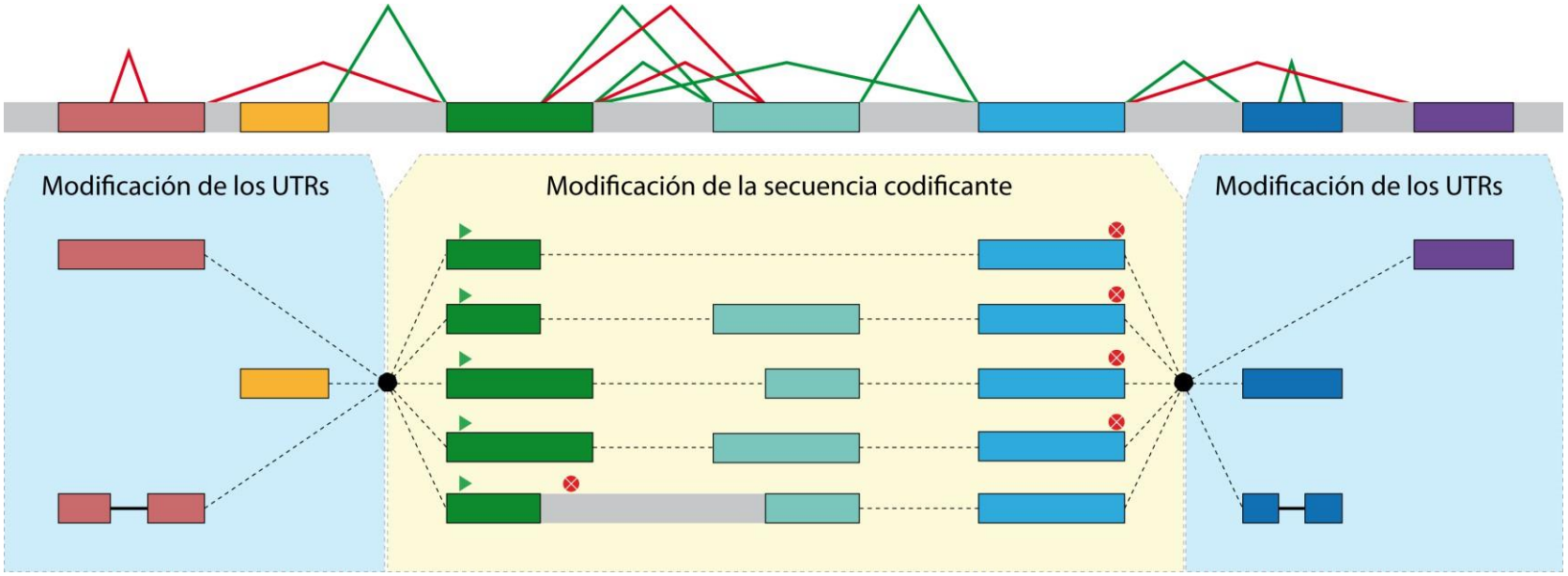
Especie	Tamaño del genoma (pb)	Nº de genes codificantes	Nº de mensajeros	Densidad génica (Nº genes/Mb)
Homo sapiens	3.324.592.091	20.769	195.565	6,2
Pan troglodytes	2.995.917.117	18.759	29.160	6,3
Gorilla gorilla	2.828.888.833	20.962	35.727	7,4
Pongo abelli	3.109.347.532	20.424	29.447	6,6
Macaca mulata	3.093.871.206	21.905	44.725	7,1
Felis catus	2.365.745.914	19.493	22.656	8,2
Canis lupus	2.392.715.236	19.856	29.884	8,3
Oryctolagus cuniculus	2.604.023.284	19.293	24.964	7,4
Rattus norvegicus	2.573.362.844	22.941	29.189	8,9
Mus musculus	3.480.528.190	23.139	93.480	6,6
Bos taurus	2.649.685.036	19.994	26.740	7,5
Gallus gallus	1.072.544.086	15.508	17.954	14,5
Danio rerio	1.505.581.940	26.247	54.869	17,4
Takifugu rubripes	393.312.790	18.523	48.706	47,1
Drosophila melanogaster	168.736.537	13.937	29.173	82,6
Arabidopsis thaliana	135.670.229	27.249	41.671	200,8
Caenorhabditis elegans	103.022.290	20.532	57.844	199,3
Saccharomyces cerevisiae	12.157.105	6.692	7.126	550,5
Escherichia coli	4.907.865	5.086	5.193	1036,3
Bacillus subtilis	4.215.606	4.185	4.371	992,7
Haemophilus influenzae	1.830.138	1.709	1.745	933,8

Ensembl: http://www.ensembl.org/index.html, **ncbi:** http://www.ncbi.nlm.nih.gov/

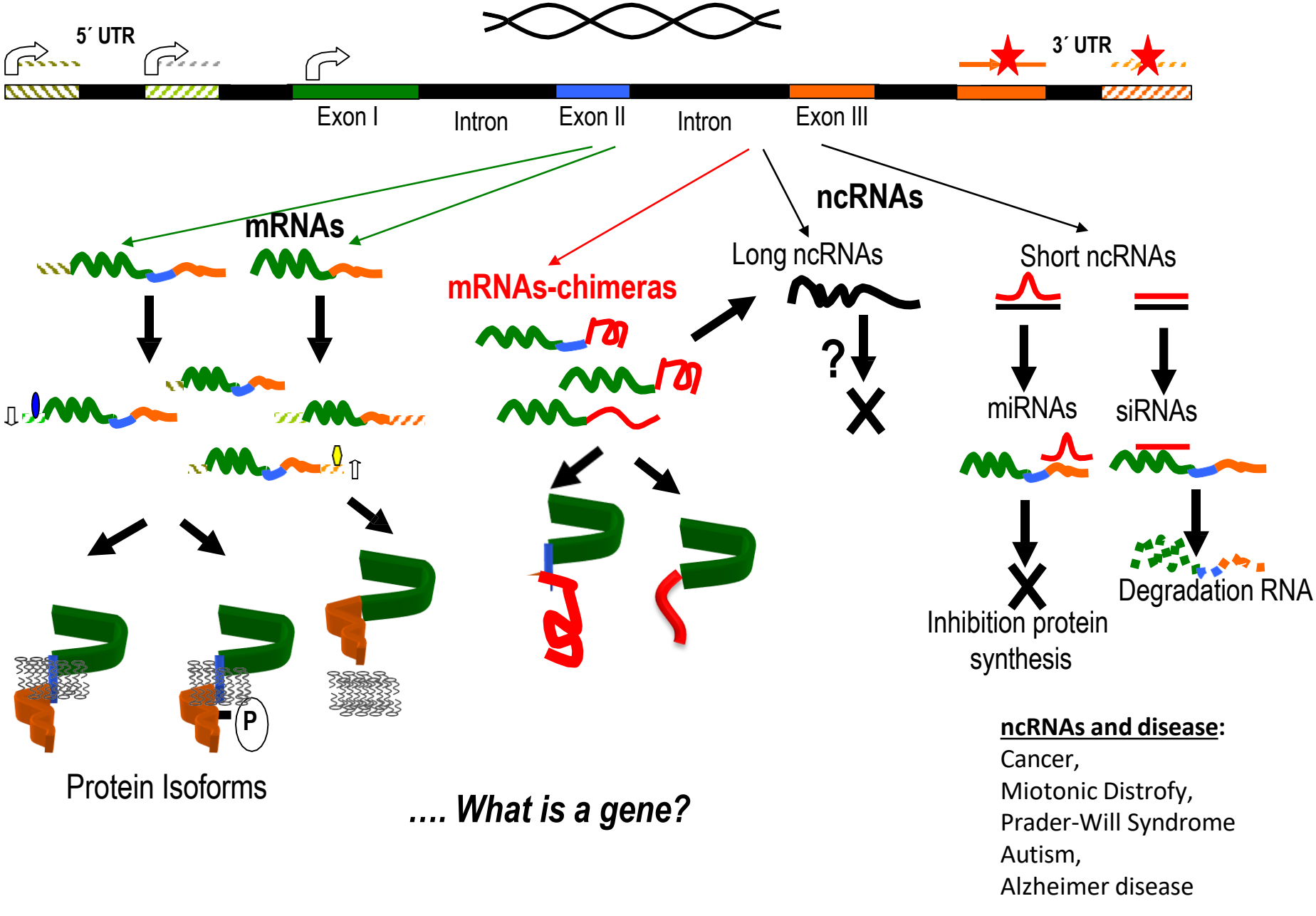
Transcripción



Splicing Alternativo



Diversidad Transcriptómica



…. What is a gene?

ncRNAs and disease:
Cancer,
Miotonic Distrofy,
Prader-Will Syndrome
Autism,
Alzheimer disease

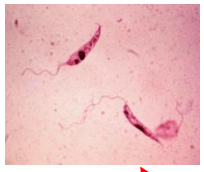
GENCODE: estadísticas en humano

Total Nº of Genes	**60.649**
Protein-coding genes	19.955
Long non-coding RNA genes	17.944
Small non-coding RNA genes	7.567
Pseudogenes	14.773
- processed pseudogenes	10.667
- unprocessed pseudogenes	3.565
- unitary pseudogenes	241
- polymorphic pseudogenes	49
- pseudogenes	15
Immunoglobulin/T-cell receptor gene segments	
- protein coding segments	409
- pseudogenes	236

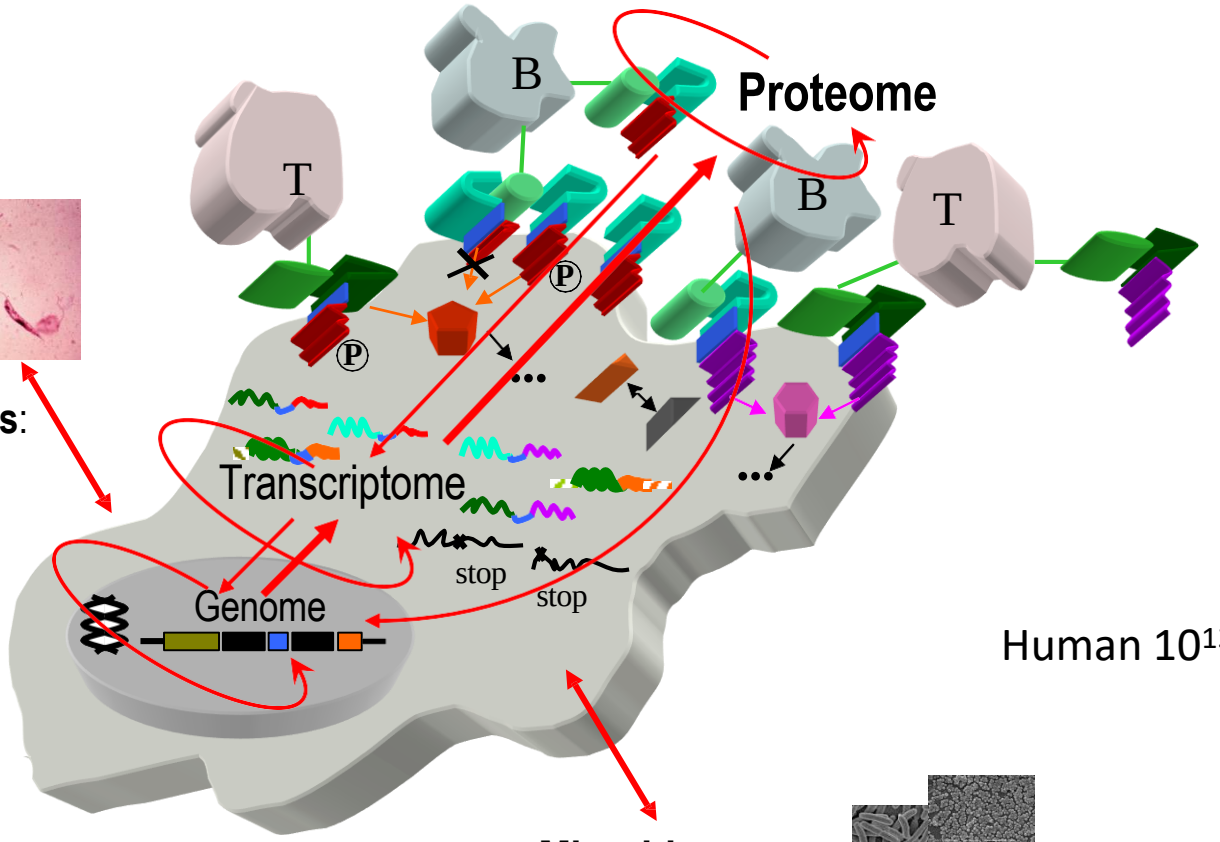
Total Nº of Transcripts	**237.012**
Protein-coding transcripts	86.757
- full length protein-coding	61.015
- partial length protein-coding	25.742
Nonsense mediated decay transcripts	18.881
Long non-coding RNA loci transcripts	48.752
Total Nº of distinct translations	63.968
Genes that have more than one distinct translations	13.689

© 2020 EMBL-EBI

Diferentes'-omas' en la célula

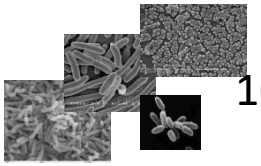


Pathogens:
Protozoa
Virus
Bacteria

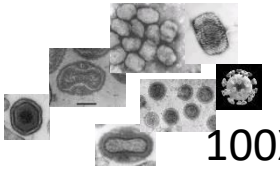


Human 10^{13} cells

Microbiome



10X bacteria



100X virus

➢1 million genes of microorganisms

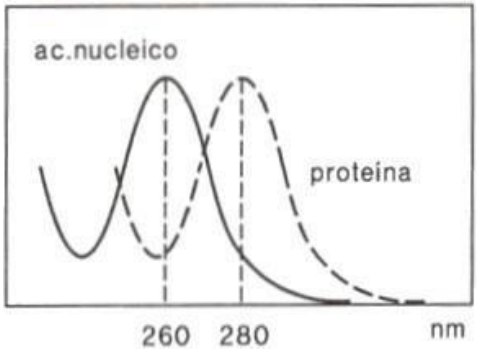
needed ⟷ disease

Nucleic Acids studies

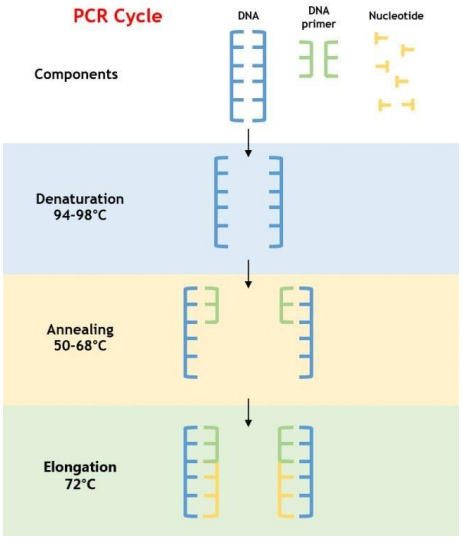
Extraction



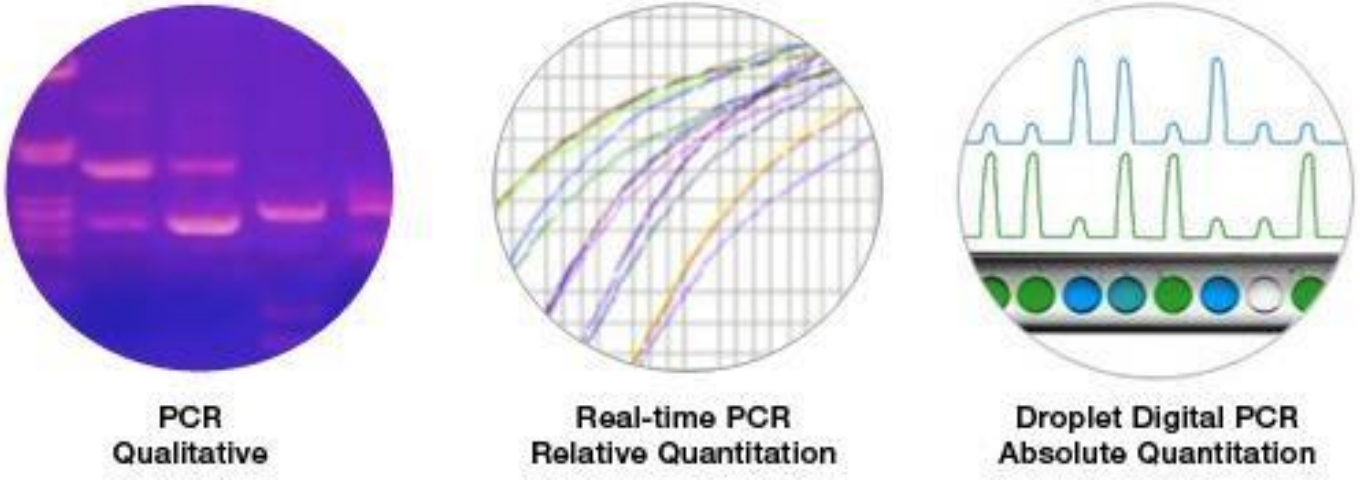
Quantification

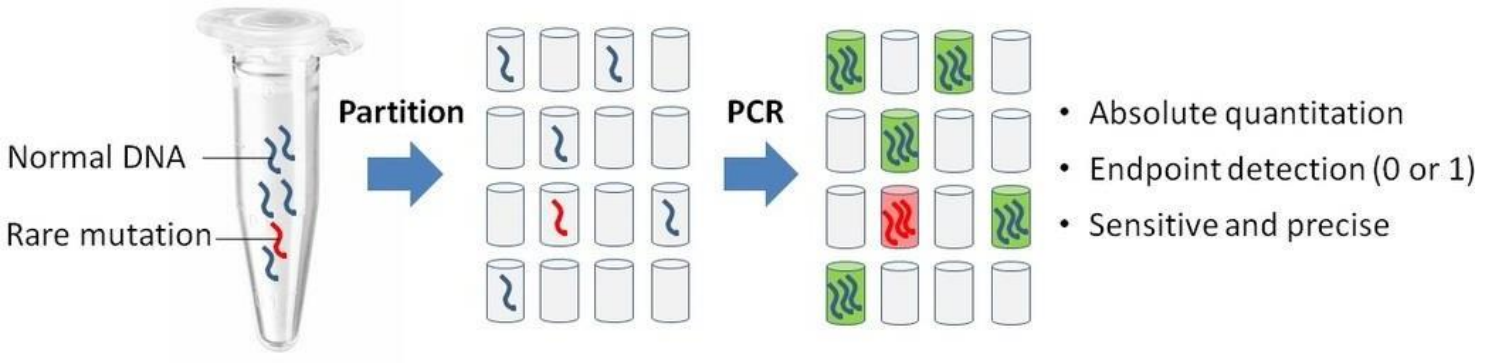


Amplification



Polymerase Chain Reaction (PCR)



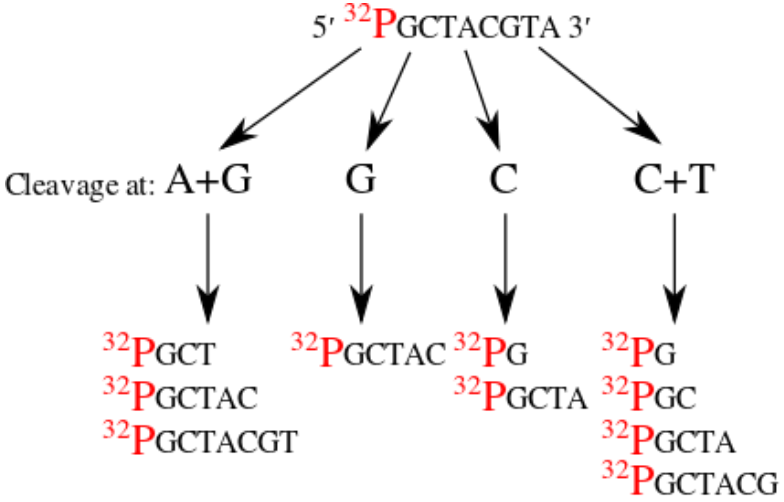


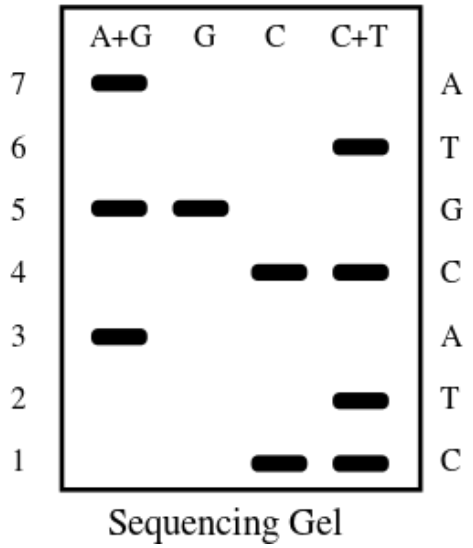
Applications: detection of rare mutants (leukemias)

RT-qPCR COVID-19

First sequencing technologies (1976-1977)

Maxam and Gilbert method
Chemical degradation of radiolabelled DNA

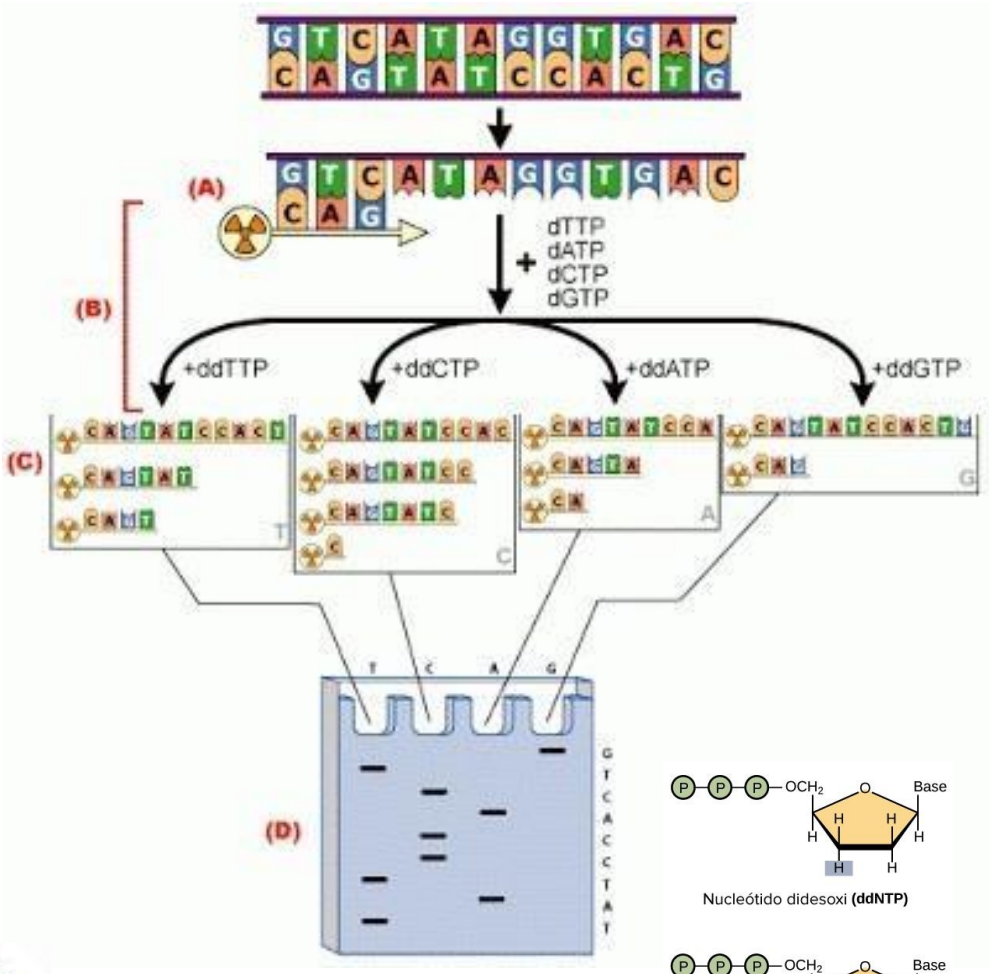




Sequencing Gel

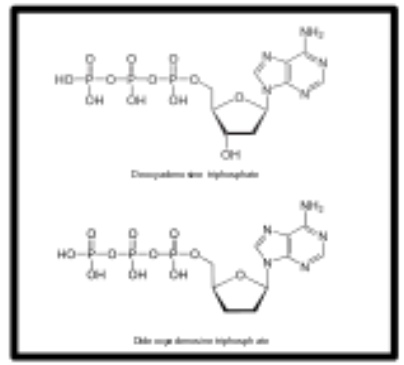


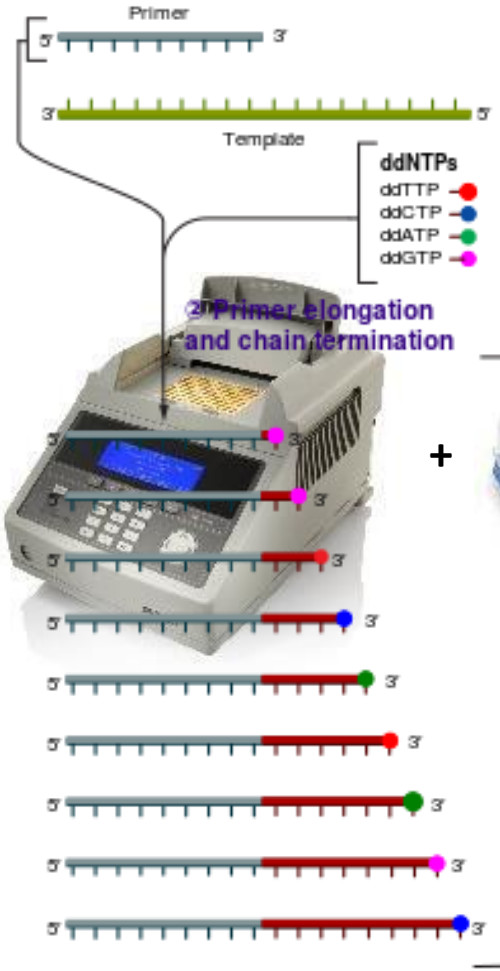
Sanger method
Synthesis and terminator



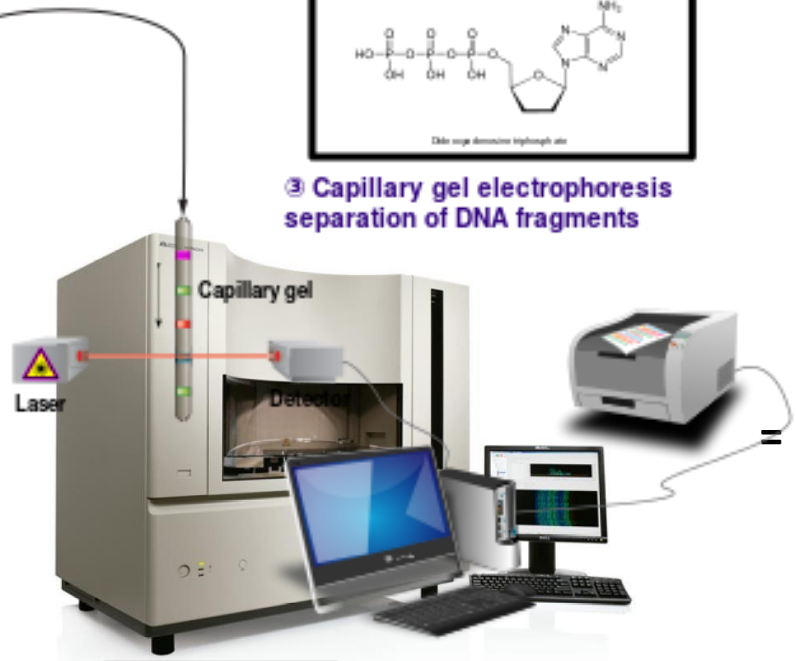
Authomatic method of Sanger (1987-1998)

① Reaction mixture
- Primer and DNA template
- DNA polymerase
- ddNTPs with flourochromes
- dNTPs (dATP, dCTP, dGTP, and dTTP)

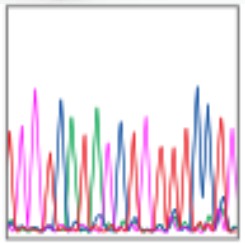








③ Capillary gel electrophoresis separation of DNA fragments



④ Laser detection of flourochromes and computational sequence analysis

PCR plate purification capilar electrophoresis and fluorescent analysis

Enzimatic method of Sanger

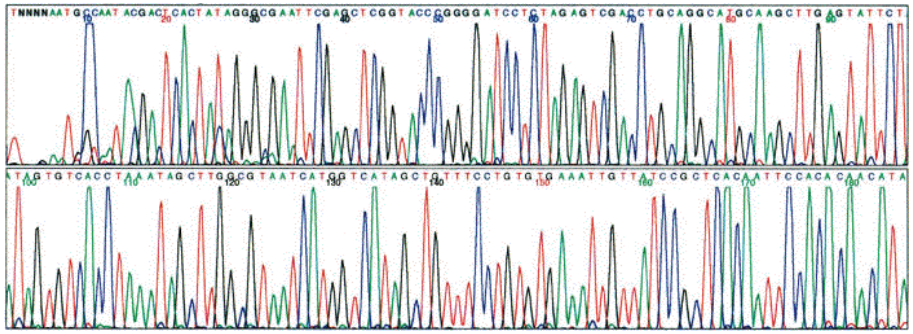
Aplicaciones:

- Confirmación de mutantes y clonajes obtenidos en el laboratorio
- Secuenciación de productos de PCR y RT-PCR
- Genotipado
- Detección de mutaciones de interés clínico en pacientes

Limitaciones:

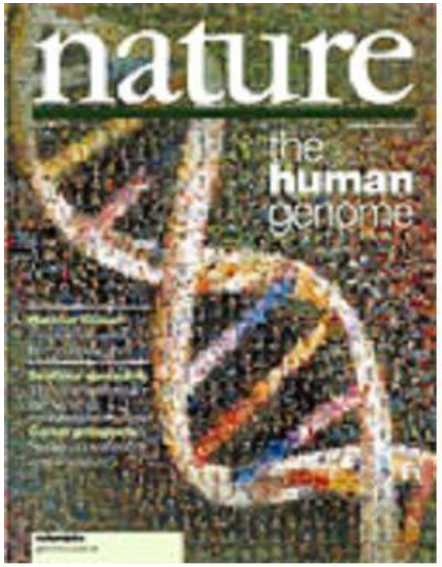
- Productividad limitada (96 secuencias por carrera)
- Coste relativamente elevado por nucleótido secuenciado
- Los proyectos sobre genomas o transcriptomas completos resultan extremadamente caros

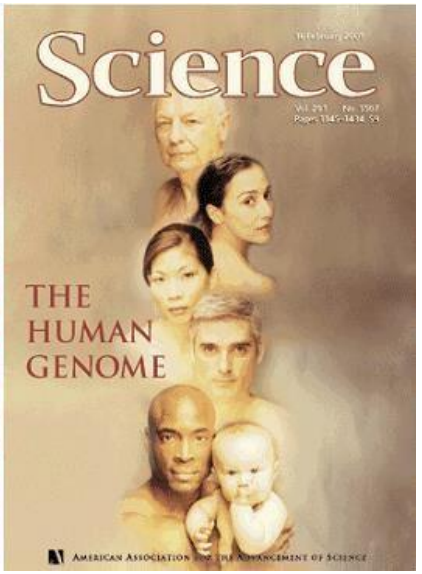
1 sample→ 1 seq **(800 nt aprox.)**



Human Genome Project 2001-2003







3.000 M bases
10 years and 3.000 M euros

Generated:
800.000 M euros
300.000 jobs

NEXT-GENERATION SEQUENCING (NGS) (2004)

Características comunes al conjunto de tecnologías:

- Muestra de entrada: **DNA / RNA (cDNA)**

- **Preparación de genoteca (librería):** fragmentación del ácido nucleico y ligación de adaptadores universales. En algunos casos, amplificación de la genoteca.

- **Secuenciación paralelizada** de un gran número de moléculas

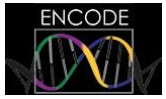
- Resultado de salida: **enorme número de secuencias** (desde decenas de miles hasta unos miles de millones por muestra y experimento)

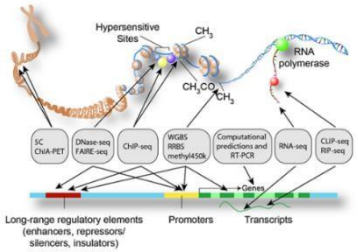
- **Bajo coste** económico por nucleótido

- Elevado coste de **análisis informático** (cuello de botella)

Human Genome Projects

ENCODE





The **ENCODE** (Encyclopedia of DNA Elements) is an international ongoing (2013) collaborative research project, aimed at identifying all the **functional elements** in the human and mouse genomes, including elements that act at the protein and RNA levels, and regulatory elements that control cells and circumstances in which a gene is active. **https://www.encodeproject.org/**

1.000 GENOMES to provide a comprehensive resource on human genetic variation, to find most genetic variants that have frequencies of at least 1% in the populations studied.
In total (2008-2015) 2500 genomes from different populations. GRCh38
IGSR: The International Genome Sample Resource (>3900 samples phase3) to ensure the future usability and accessibility of data from the 1000 Genomes Project and to extend the data set to include new data**. http://www.1000genomes.org**

100.000 GENOMES (GENOMICS ENGLAND).
Combining genomic sequence data with medical records. 70.000 patients and families **rare diseases and cancers (39.540). http://www.genomicsengland.co.uk/**

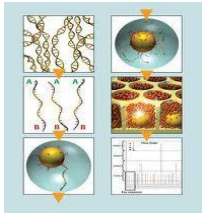
1.000.000 GENOMES (All of Us) Precision Medicine Initiative. USA.
Disease prevention and treatment that takes into account individual variability in enviroment, lifestile and genes of each person. The program would focus not just on disease, but also in increase chances to remain healthy.

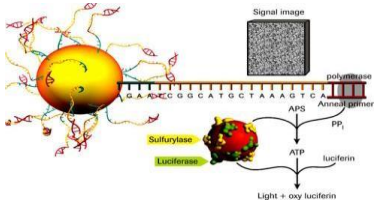
4P medicine. (**Predictive, Preventive, Personalised/precise, Participative**)
"Human Genomics Projects and precisión medicine" Gene Therapy 2017
Carrasco-Ramiro F, Peiró-Pastor R, **Aguado** B.

Second and third generation sequencers (2004-)

SECOND GENERATION

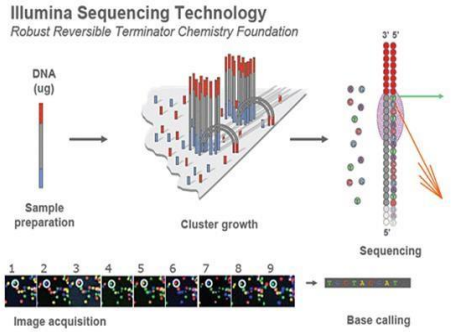
ROCHE-454, pirosequencing



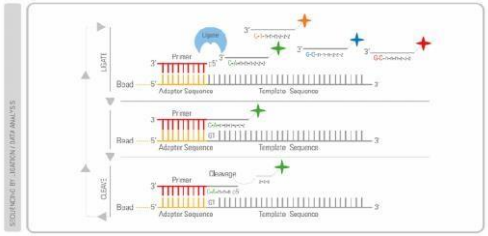




ILLUMINA, terminators and bridge PCR

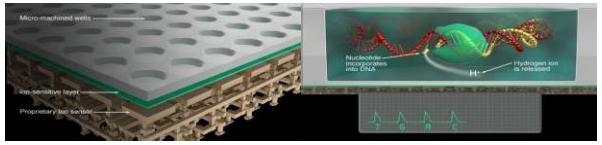


SOLiD Sequencing by ligation





LIFE TECH-THERMO, semiconductors (pH)

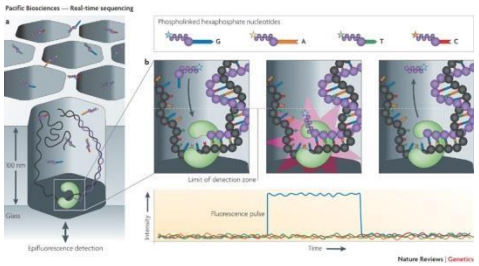




THIRD GENERATION (single molecule)

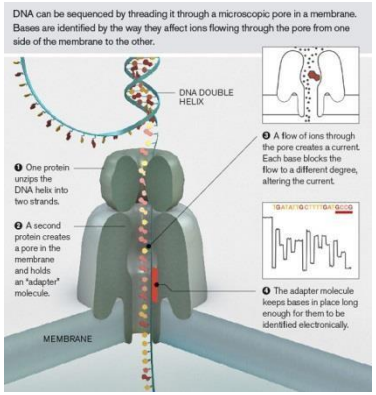
PACIFIC BIOSCIENCES

Synthesis *zero-mode waveguides*

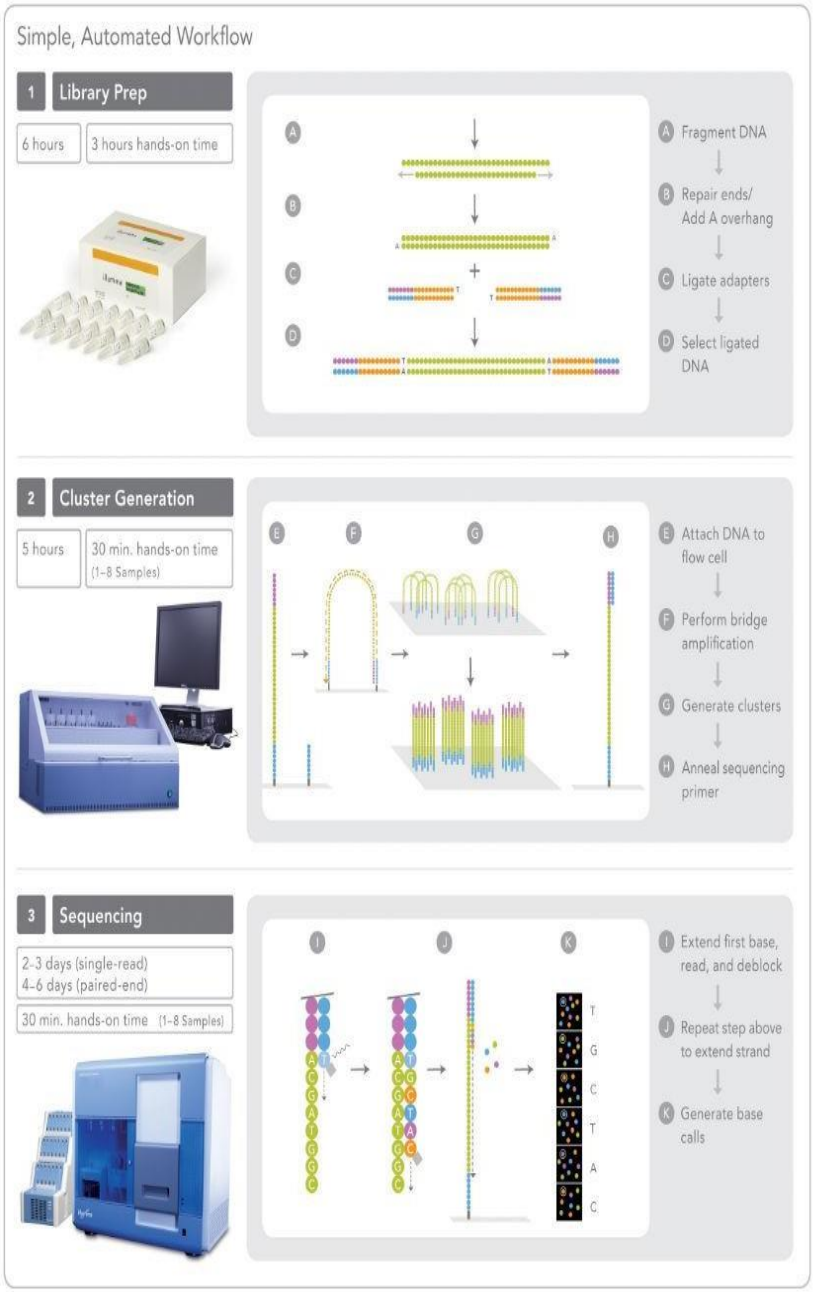


OXFORD NANOPORE,

interruption electric current



ILLUMINA (“short reads” sequencing)



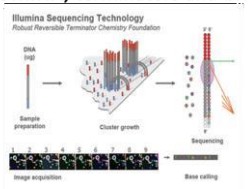
“terminators and bridge PCR”

https://youtu.be/fCd6B5HRaZ8

1 cluster - 1 read (seq)

ILLUMINA (DOMINANT POSITION)

ILLUMINA, terminators and bridge PCR



terminators and bridge PCR

- Up to **52.000 million** reads per run (16 Tbases)

- Single (1x35 nt, 1x50, …1x300)
- Paired-end (2x35, 2x50, …2x300)
- Mate-pair (2x35, inserts of 2, 5 and 10 kb)

- **Applications:**
- Resequencing
- Transcriptomics (gene expression and smallRNAs)
- ChIP-seq
- Epigenomics
- Amplicons



Novaseq X Series (26.000M-52.000M reads 2x150)



Novaseq 6000 (10-20.000M reads 2x250)

NextSeq 1000 & 2000 (1.800M reads 2x300)



NextSeq 550 (400M reads 2x150)



MiSeq (25M reads 2x300)

MiniSeq (25M reads 2x150)

iSeq 100 (4M reads 2x150)

Production-scale

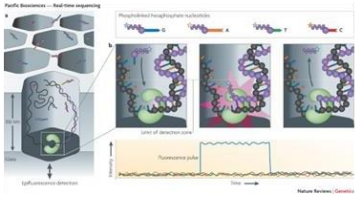
Bench Top

https://emea.illumina.com/systems/sequencing-platforms.html

PACIFIC BIOSCIENCES (“long reads” seqs)

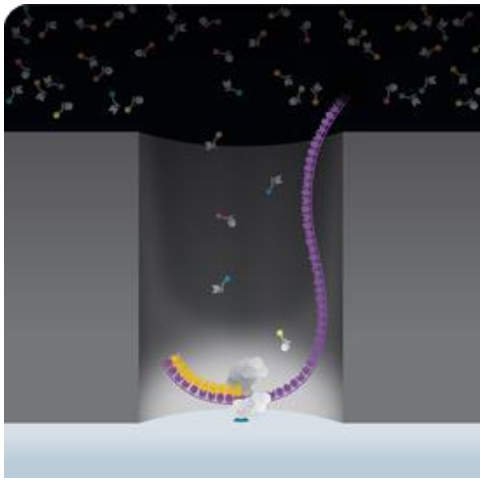
PACIFIC BIOSCIENCES

Synthesis *zero-mode waveguides*



SMRT technology:

- **library prep free of amplification**
- **Sequencing** by synthesis in *zero-mode waveguides* **(single molecule)**



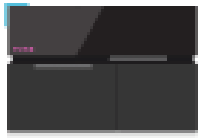
Circular Consensus Sequencing (CCS): The SMRTbell DNA template allows polymerase to sequence the same DNA molecule multiple times with multiple passes producing high intra-molecular consensus accuracy. **Phi29 polymerase**

SEQUEL IIe



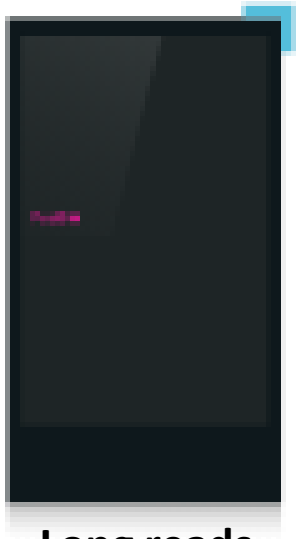
Long reads

ONSO



Short reads

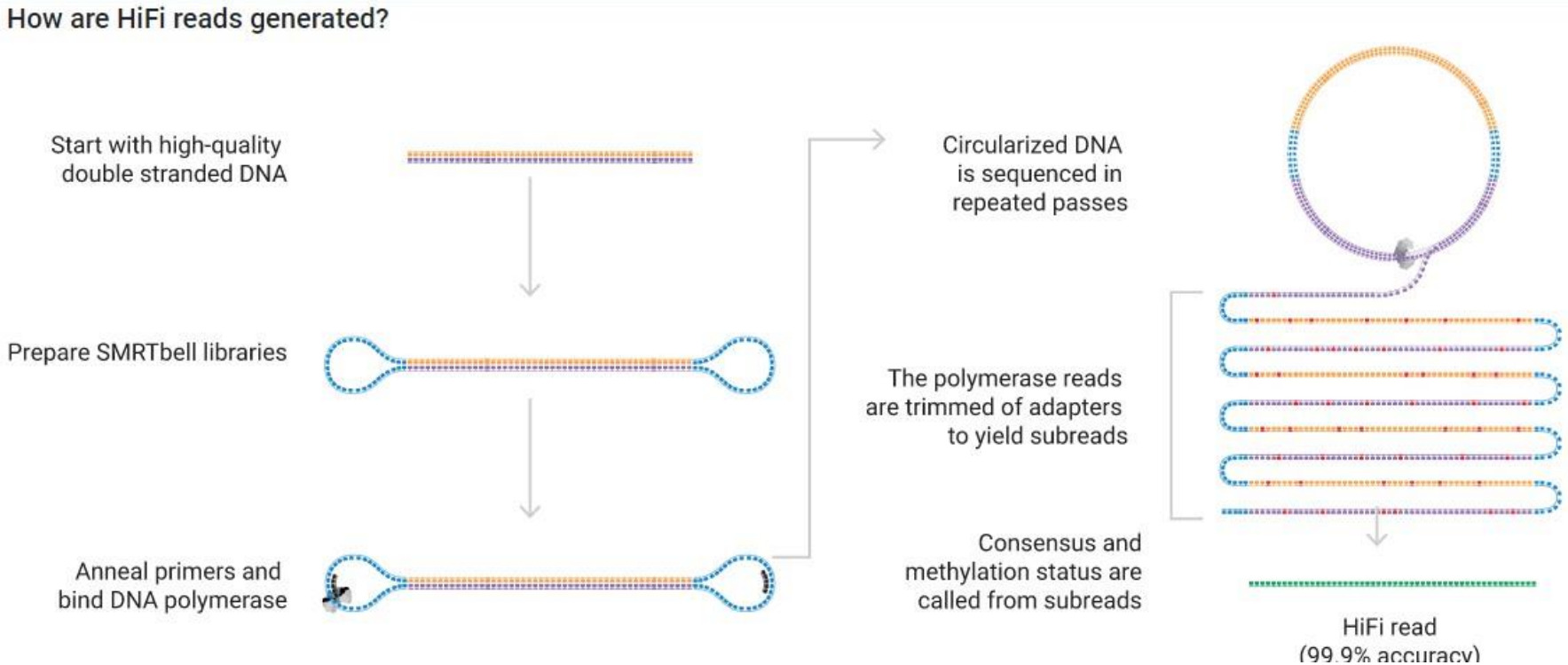
REVIO



Long reads

https://www.youtube.com/watch?v=v8p4ph2MAvI

HIFI READS FOR HIGHLY ACCURATE LONG-READ SEQUENCING



SEQUEL IIe



- SMRT **Cell 8M**
- **Accurate HiFi** Long reads (**15-20.000 pb**)
- 5-base genome sequencing (**A, T, G, C, 5mC**) in native DNA

- **Applications:**
  - *de novo* sequencing (small and large genomes)
  - re-sequencing
  - Transcriptomics (complete transcripts)
  - Modified bases

https://youtu.be/_lD8JyAbwEo

Sequence with HiFi reads at high throughput

	Sequel® IIe system	Revio system	Increase
Higher density	8 million ZMWs	25 million ZMWs	3×
Independent stages	1	4	4×
Shorter run times	30 hours	24 hours	1.25×
30× HiFi human genomes/year	88	1,300	15× overall

REVIO



(2023)

The $1,000 complete, phased genome

HiFi sequencing provides small variants, structural variants, repeat expansions, methylation, and haplotype phasing from a single library and sequencing run

SBB CHEMISTRY, SEQUENCING BY BINDING

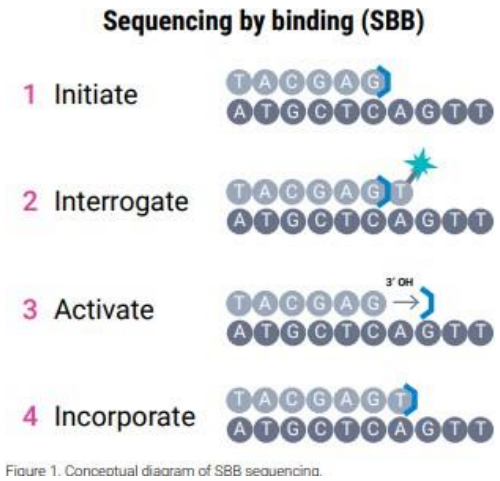
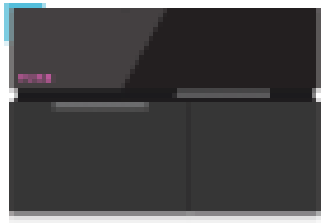


Figure 1. Conceptual diagram of SBB sequencing.

ONSO



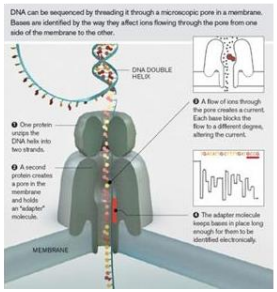
SPECIFICATIONS

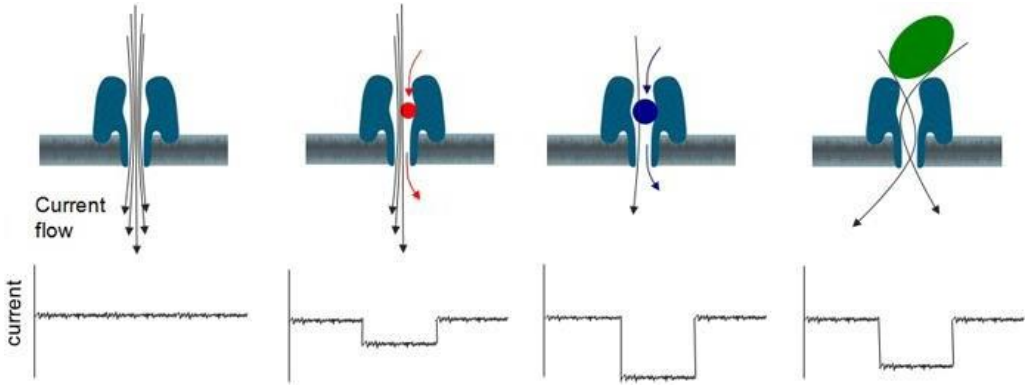
Onso reagents	Read length	Reads	Output (Gb)	Run time	Quality score
200 cycle sequencing kit	1x200 bp 2x100 bp	400 – 500M (SE) 800 – 1,000M (PE)	80 – 100	32 hours	≥90% Q40
300 cycle sequencing kit	2x150 bp	800 – 1,000M (PE)	120 – 150	48 hours	≥90% Q40

https://youtu.be/i_mSaNBOVmQ

OXFORD NANOPORE (“long reads”)

OXFORD NANOPORE,
interruption electric current





Sequencing by **nanopors:** interruption of electric current by the sample



MinION **FROM $1,999**

The only portable, real-time devices for DNA and RNA sequencing, giving complete control and creativity over when, where and how often you sequence, regardless of application.



GridION **FROM $67,000**

Compact benchtop device designed to run and analyse up to five MinION or Flongle Flow Cells



PromethION **FROM $9,555**

High-coverage nanopore sequencing in formats ranging from modular, fully-integrated devices, to high-throughput solutions. Each flow cell can deliver the lowest price per Gb for nanopore sequencing.

https://nanoporetech.com/support/how-it-works#fullVideo



Configuration	Platform			Techniques	Tech specifications			
Number of flow cells per device	1	1	1	5	2	2	24	48
Maximum number of channels per flow cell	512	512	512	512	2,675	2,675	2,675	2,675
Run time	72 Hours	72 Hours	72 Hours	72 Hours	72 Hours	72 Hours	72 Hours	72 Hours
Device TMO†	50 Gb	50 Gb	50 Gb	250 Gb	580 Gb	580 Gb	~7 Tb	[illegible] Tb

	PacBio Revio	Nanopore sequencing
Read length	15–18 kb	10–100 kb
Read accuracy	99.95% (Q33)	99.26% (Q21)
Run time	24 hours	72 hours
Yield	90 Gb	50–110 Gb
Variant calling — SNVs	✓	✓
Variant calling — indels	✓	X
Variant calling — SVs	✓	✓
5mC methylation	✓	✓
Phasing	✓	✓

NGS applications

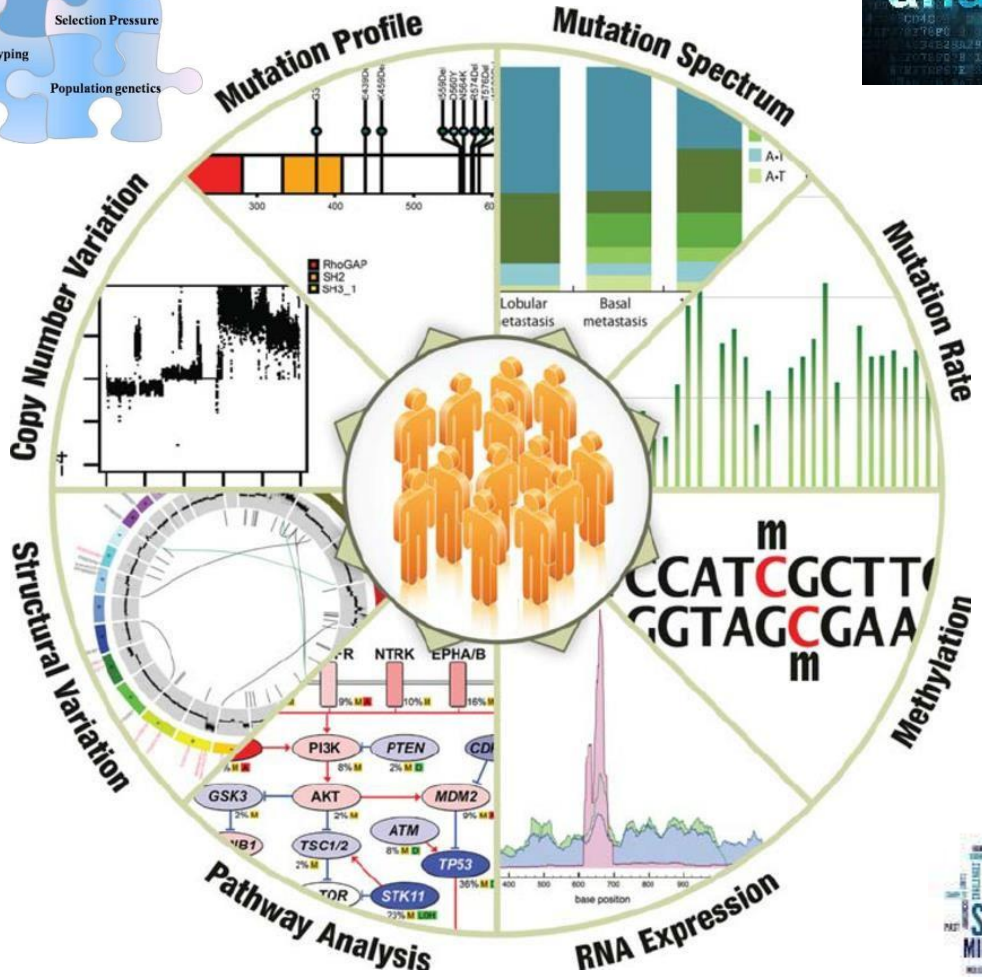
1) **Secuenciación "de novo"** y ensamblaje de genomas
2) **Resecuenciación** de genomas: búsqueda de SNPs, mutaciones puntuales, InDels, …etc.
3) **Secuenciación dirigida** a regiones específicas de interés (exomas, loci informativos para enfermedades o para clasificación taxonómica, …etc.)
4) **Transcriptomas (RNA-seq):** secuenciación "de novo" y ensamblaje
5) **Expresión génica (RNA-seq):** comparación entre dos o más condiciones experimentales a nivel de transcriptoma completo
6) **Sequenciación de amplicones (16S, 18S, ITS):** Identificación y comparación de procariotas, eucariotas o de hongos en poblaciones.
7) **ChIP-seq:** estudios interacción proteína-DNA. Secuenciación de cromatina inmunoprecipitada
8) **Epigenética** (metilación, …etc.)
9) **ATAC-seq** (Transposase-Accessible chromatin)
10) **Diseases:** diagnostic, prognostic and personal treatment
11) **Metagenómica:** estudio del conjunto de organismos vivos en una muestra (agua, suelo, sistema digestivo, …etc.)
12) **Paleogenómica**: análisis de muestras antiguas
13) **Single cell genomics/transcriptomics**
14) **Enfermedades /Medicina 4P :** diagnóstico, pronóstico y tratamiento individualizados

Computational Analisys: fundamental





…but…how?
I do not have that expertise…







“ ”



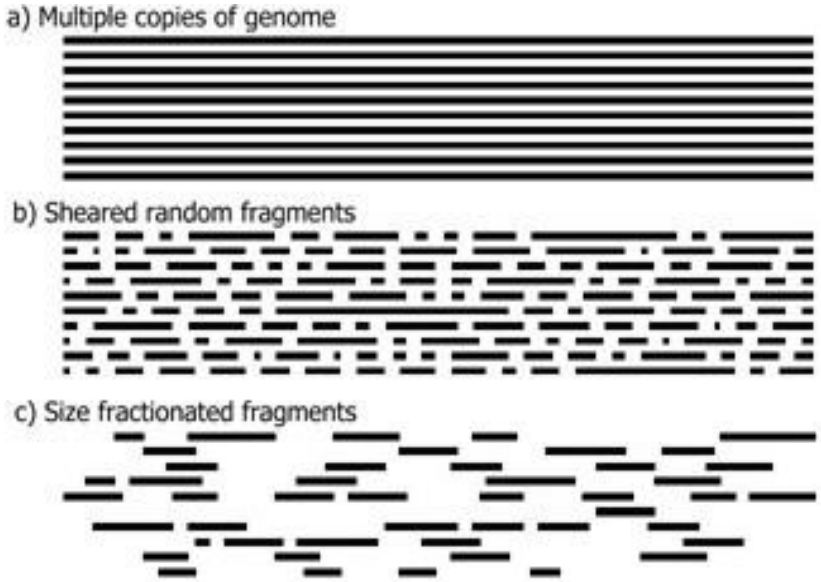
General masive sequencing data analysis:
DNAseq, RNAseq, ChipSeq, ATACseq, metagenomics, single cell, amplicons, annotation, submission to DDBB…

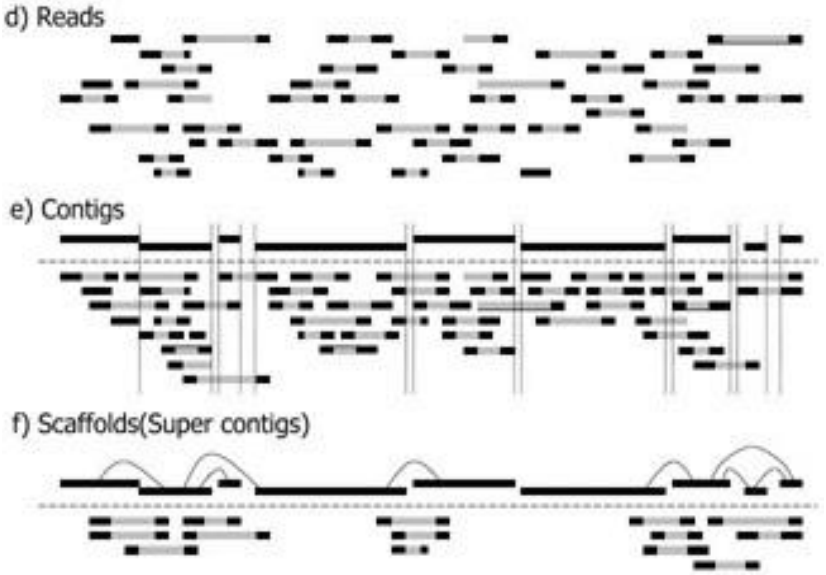
Custom analysis:
DNAseq: close/complete genome assembly, annotation and re-annotation, rare variants,
RNAseq: isoforms, annotation and re-annotation,
Amplicons: Detailed studies
Machine Learning
Data Mining
Single cell: deconvulotion, …
Omics integration: genomics+transcriptomics+epigenomics+proteomics+metabolomics….
Statistics
Graphics, Tables, …
Knowledge Discovery in Databases (KDD)
Imaging
submission to DDBB
….



NGS:- Applications and data analysis: DNAseq (de novo)

***de novo* sequencing:** genome assembly





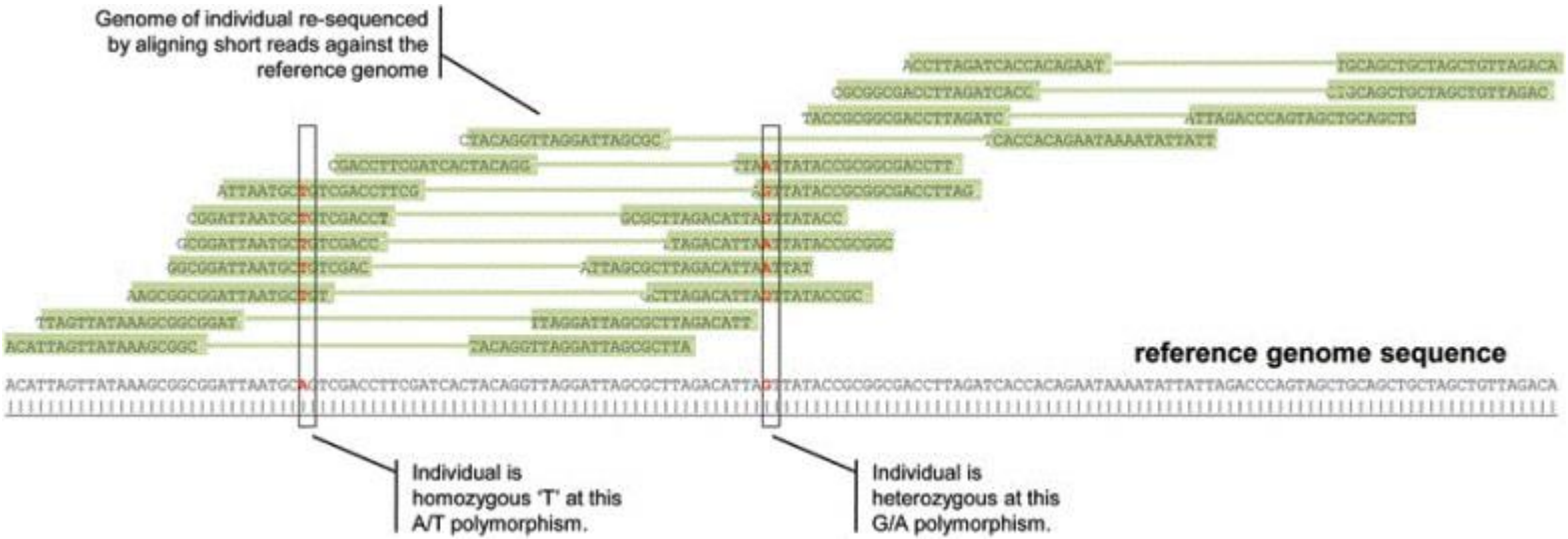
Combine short reads (Illumina) with long reads (PacBio, Oxford Nanopore)

- High coverage

NGS:- Applications and data analysis: DNA reseq

- **Genome Resequencing:** SNPs, InDels, …etc.

- **Direct sequencing:** specific regions (exomes, loci, …etc.)

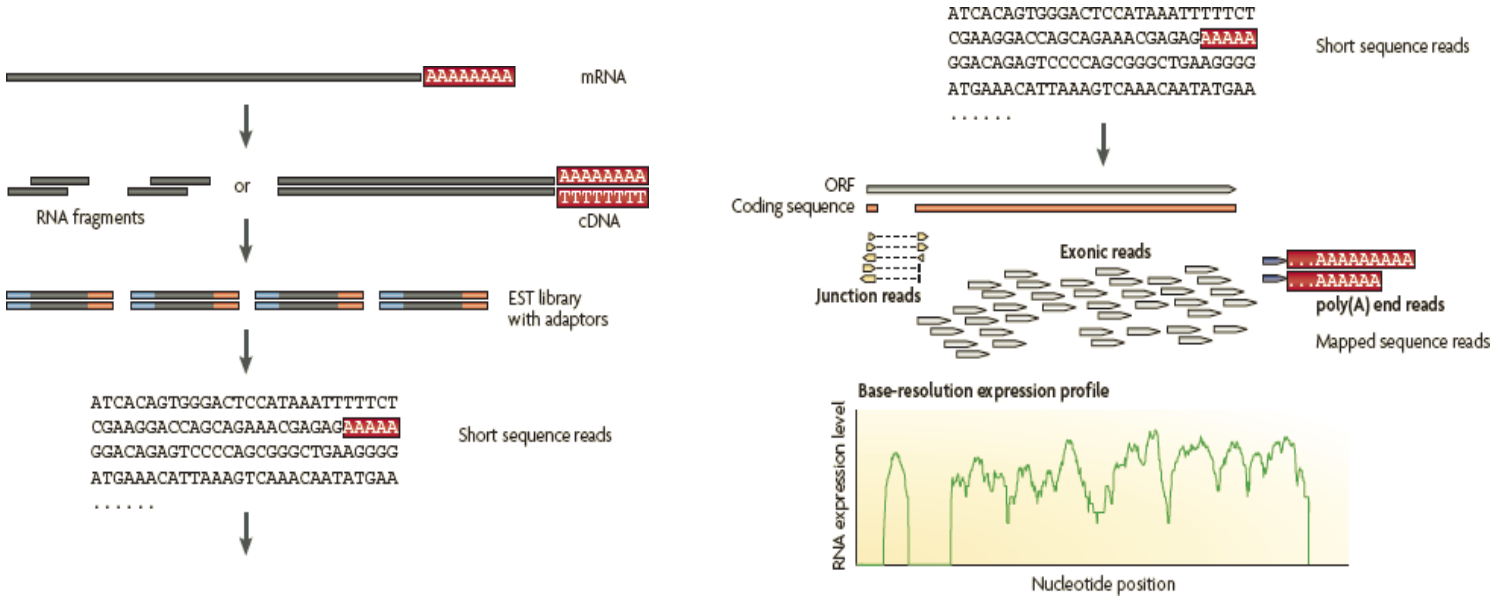


Reference genome. Short reads

- Low/high coverage

NGS:- Applications and data analysis: RNAseq

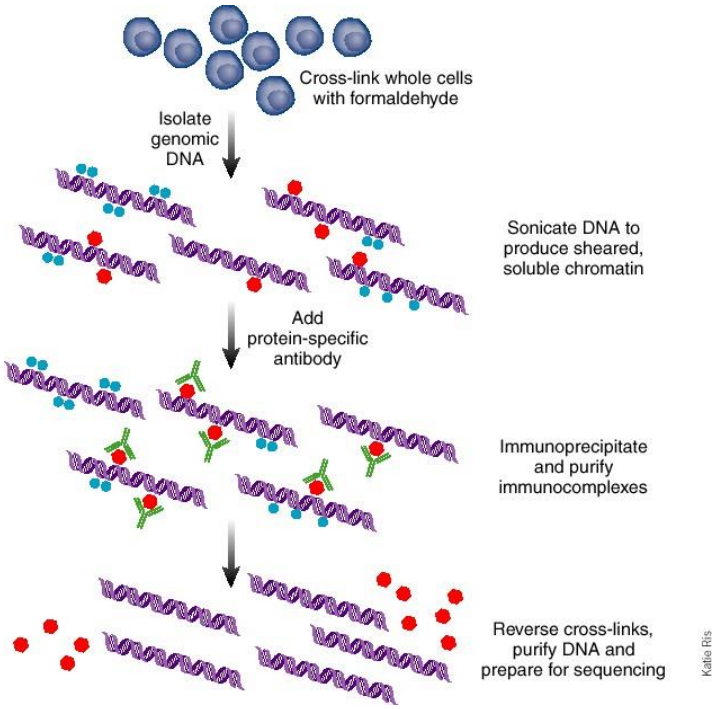
- ***de novo* sequencing** and assembly of **transcriptomes**
- **gene expression** of two or more conditions

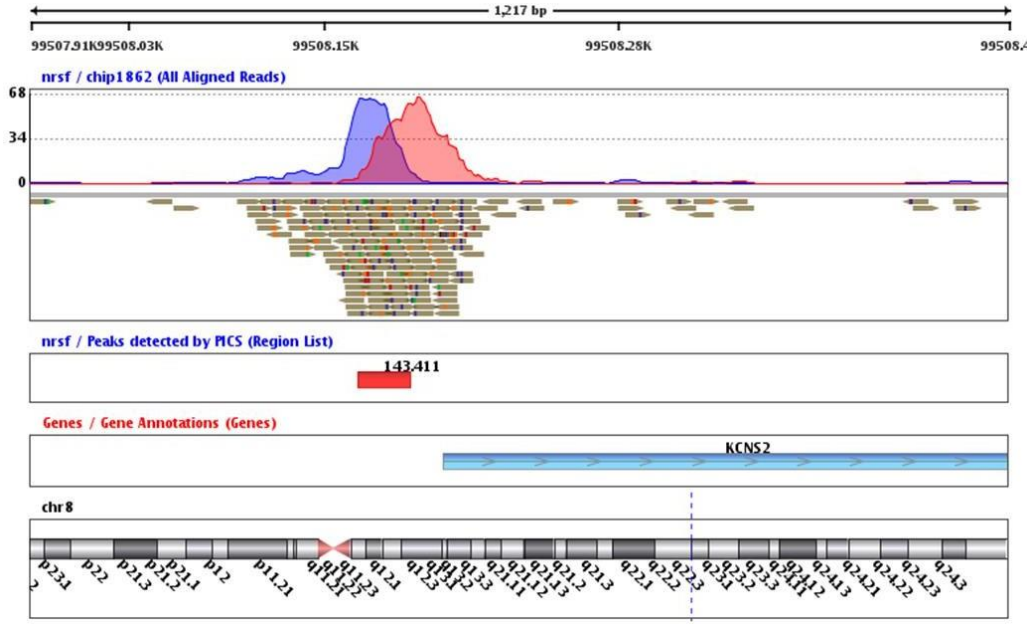


Short (Illumina) or long reads (Iso-Seq: PacBio)
Low/high coverage

NGS:- Applications and data analysis: Chip-seq

- **protein-DNA interaction**

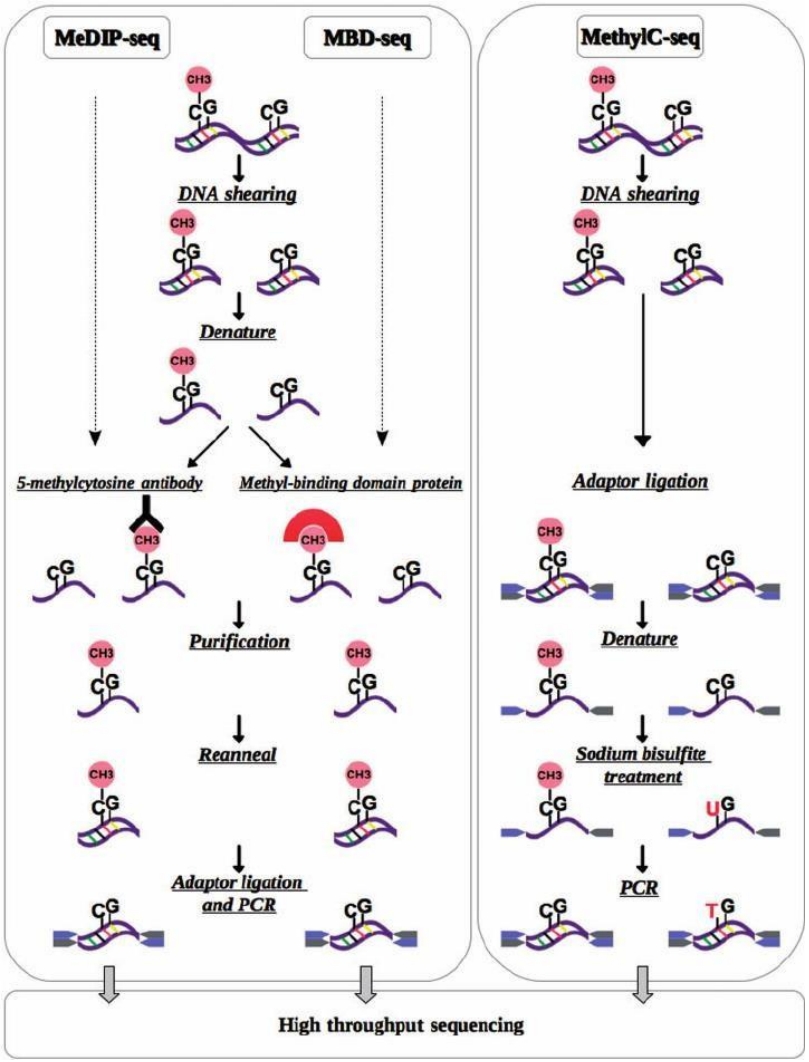


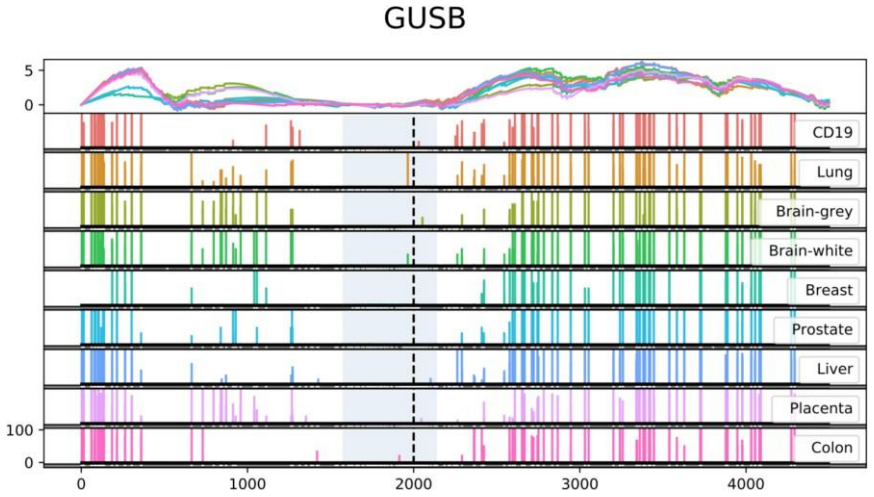


Short reads
Low/high coverage

NGS:- Applications and data analysis: Methil_DNA

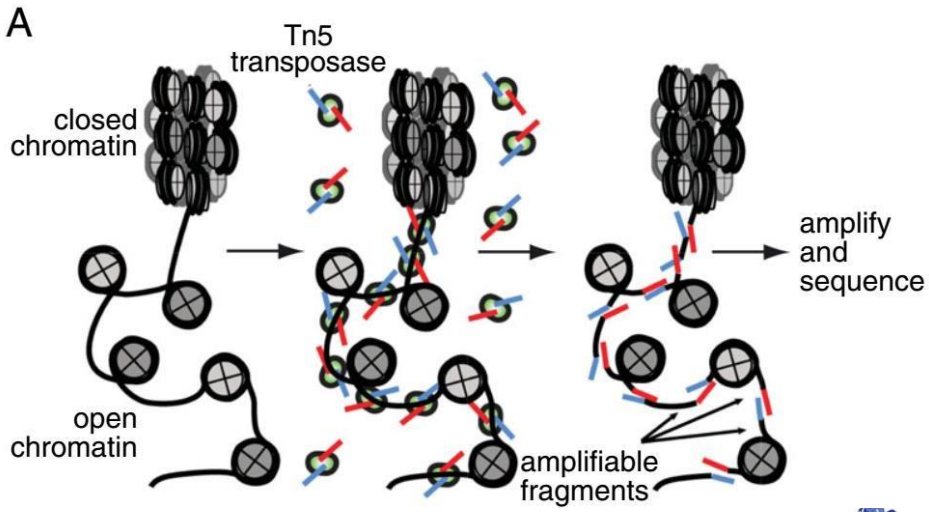
- **DNA methilation**

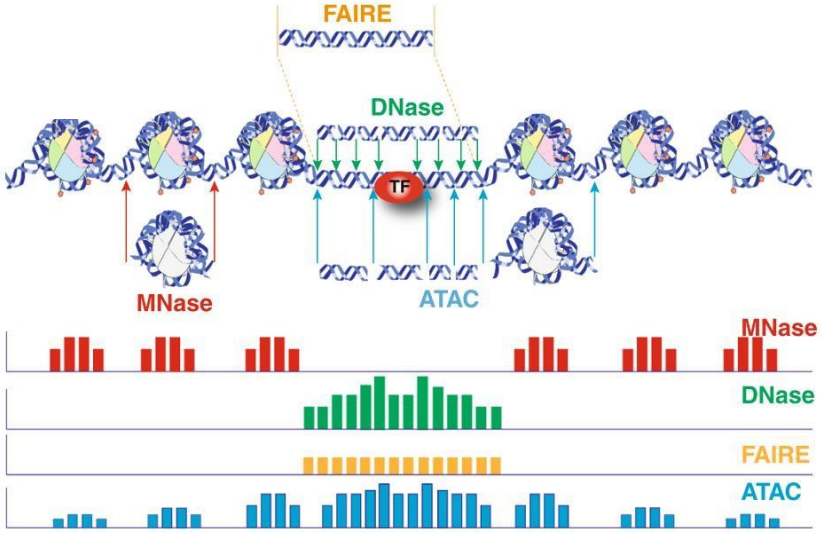




Short reads
Long reads Oxford Nanopore
Low/high coverage

NGS:- Applications and data analysis: ATAC-seq



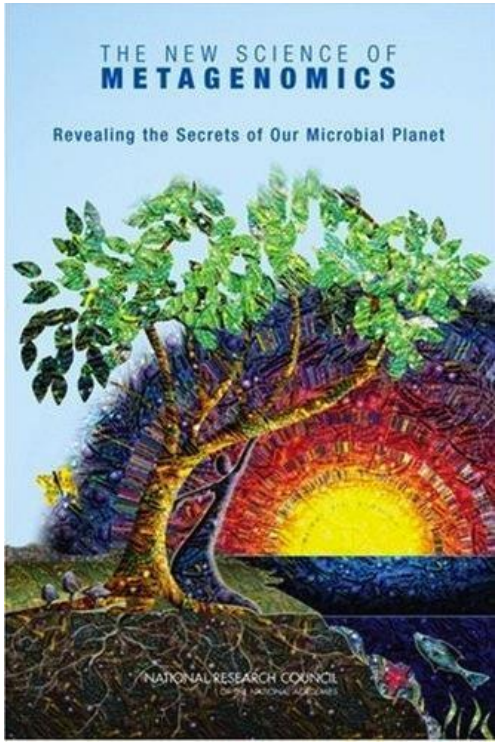


Short reads
Low/high coverage

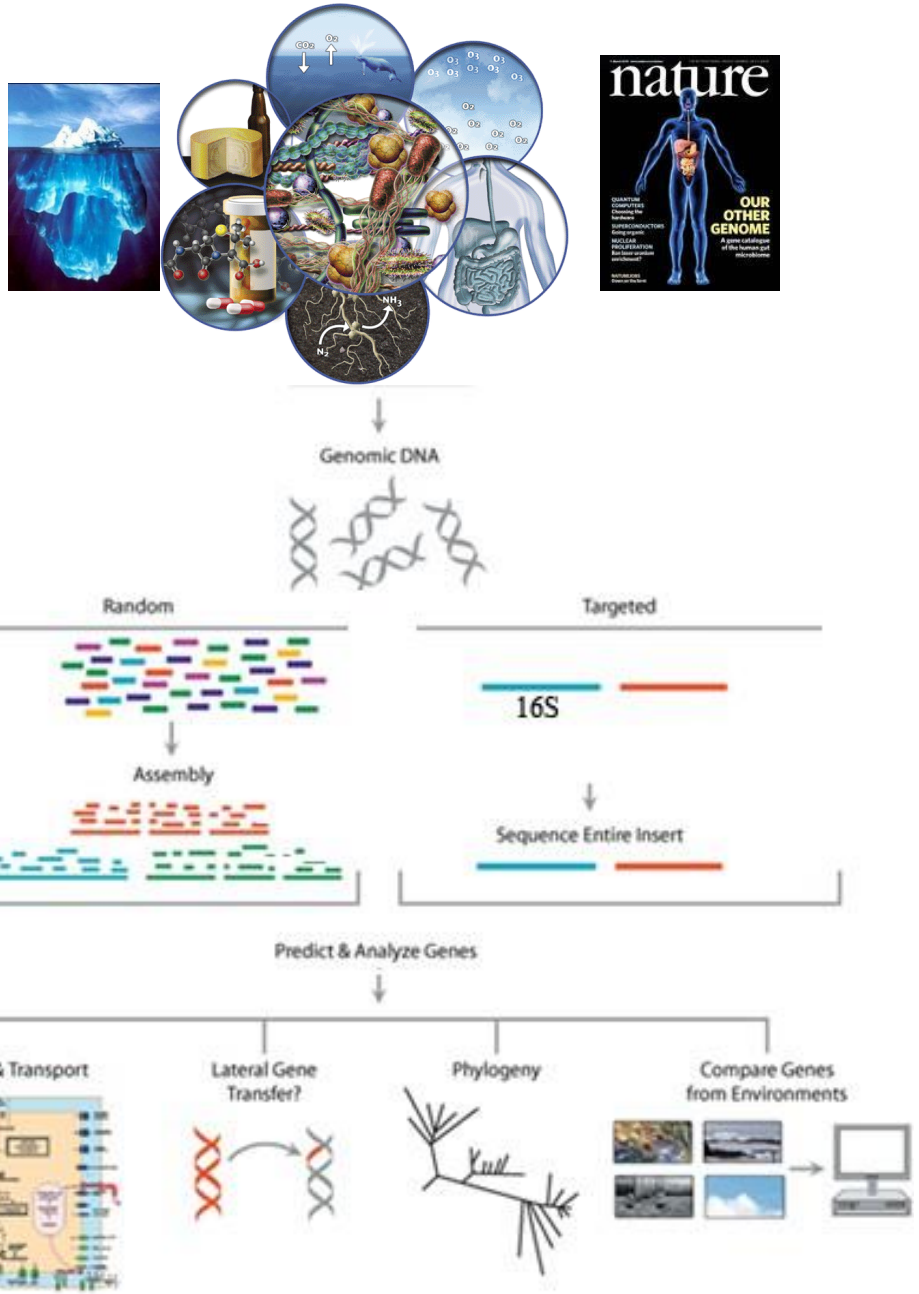
Figure 1 Schematic diagram of current chromatin accessibility assays performed with typical experimental conditions. Representative DNA fragments generated by each assay are shown, with end locations within chromatin defined by colored arrows. Bar diagrams represent data signal obtained from each assay across the entire region. The footprint created by a transcription factor (TF) is shown for ATAC-seq and DNase-seq experiments.

ATAC-seq (Assay for Transposase-Accessible Chromatin with high-throughput sequencing) is a method for determining chromatin accessibility across the genome. It utilizes a hyperactive Tn5 transposase to insert sequencing adapters into open chromatin regions.

NGS:- Applications and data analysis: Metagenomics

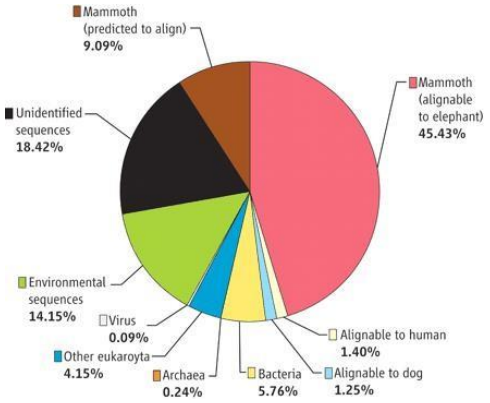


HYDROCARBON METAGENOMICS

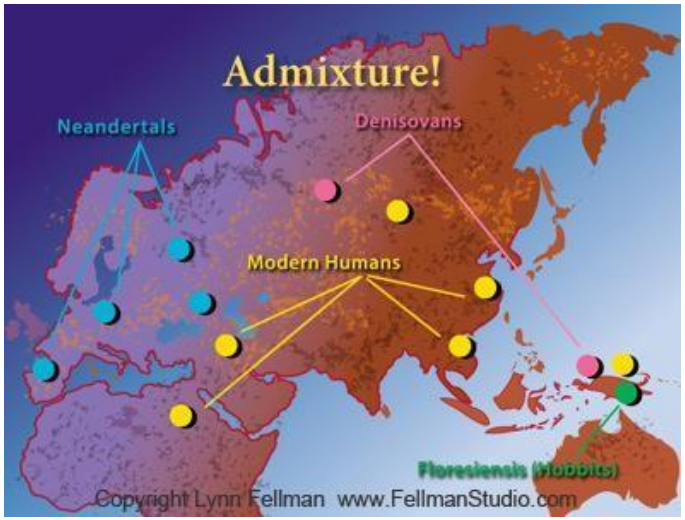


NGS:- Applications and data analysis: Paleogenomics

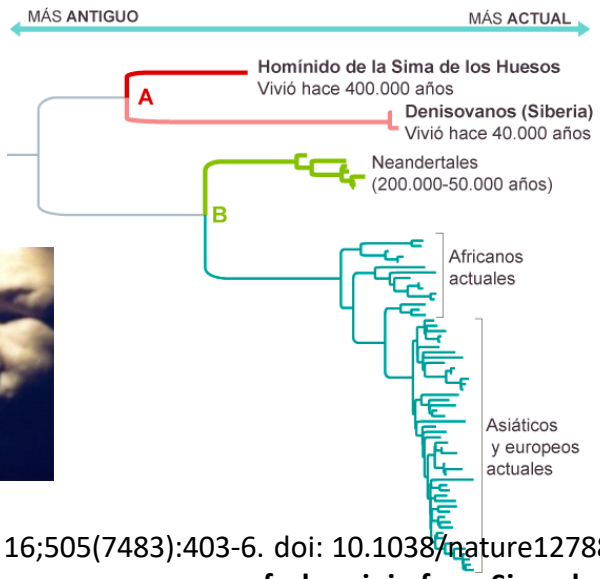
Genome of extinguished species : The mammoth







Our ancestros



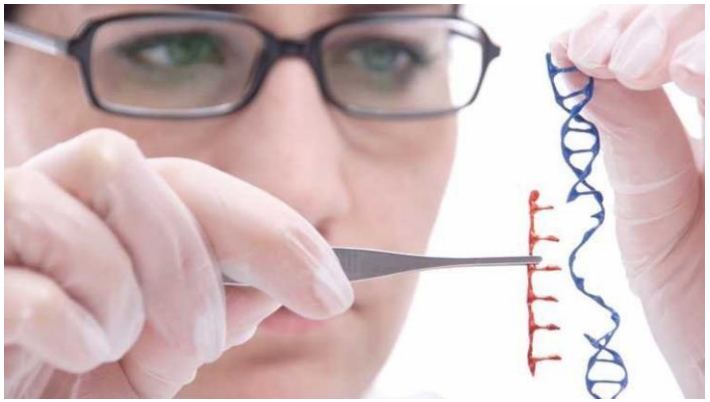




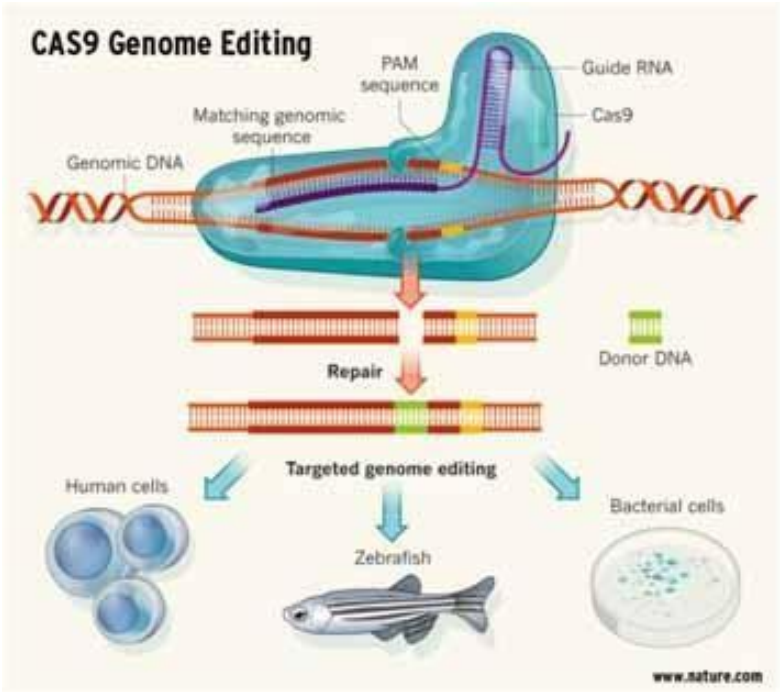
Nature. 2014 Jan 16;505(7483):403-6. doi: 10.1038/nature12788. Epub 2013 Dec 4.
A mitochondrial genome sequence of a hominin from Sima de los Huesos.
Meyer M[1], Fu Q[2], Aximu-Petri A[1], Glocke I[1], Nickel B[1], Arsuaga JL[3], Martínez I[4], Gracia A[4], de Castro JM[5], Carbonell E[6], Pääbo S[1].

NGS:- Applications and data analysis: Genome Editing



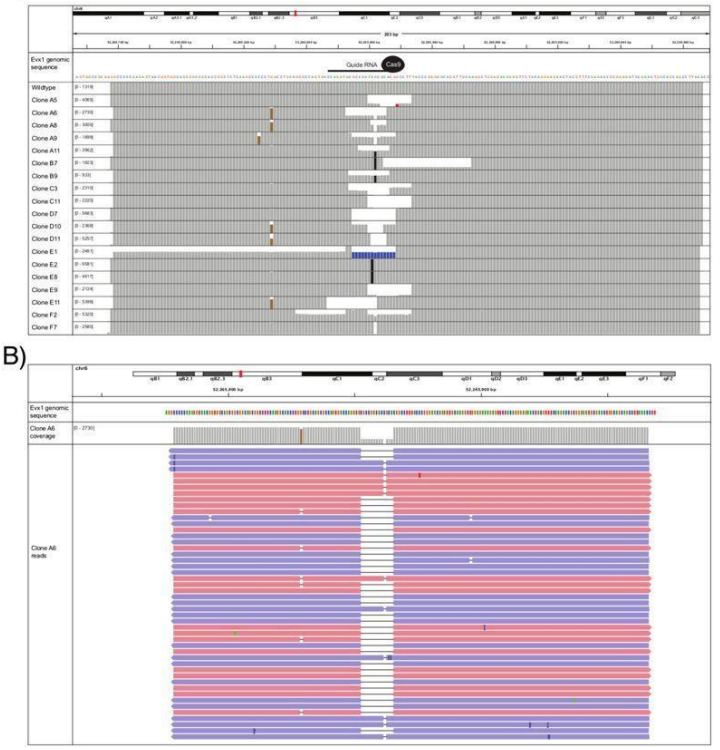


CRISPR/Cas9 and Targeted Genome Editing: A New Era in Molecular Biology

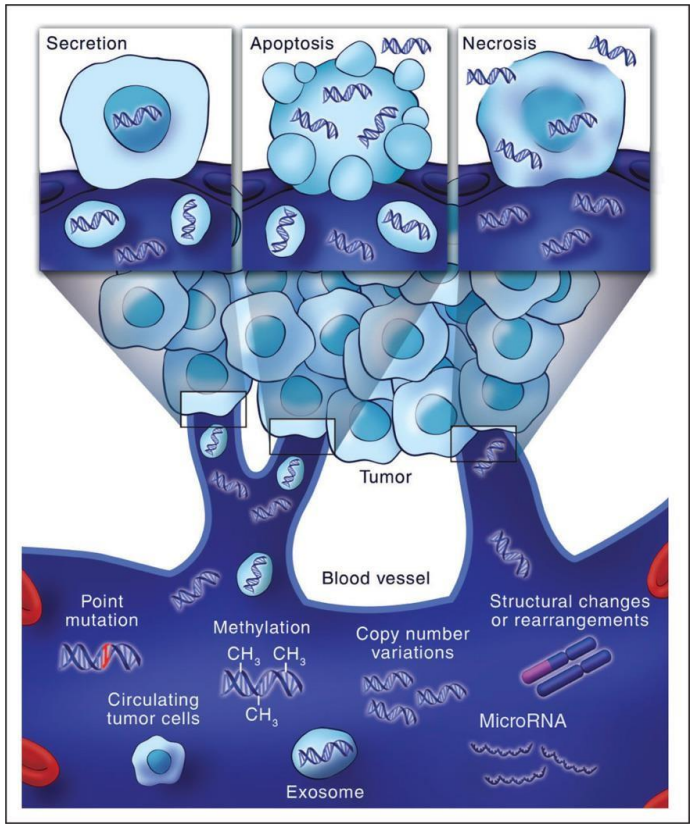


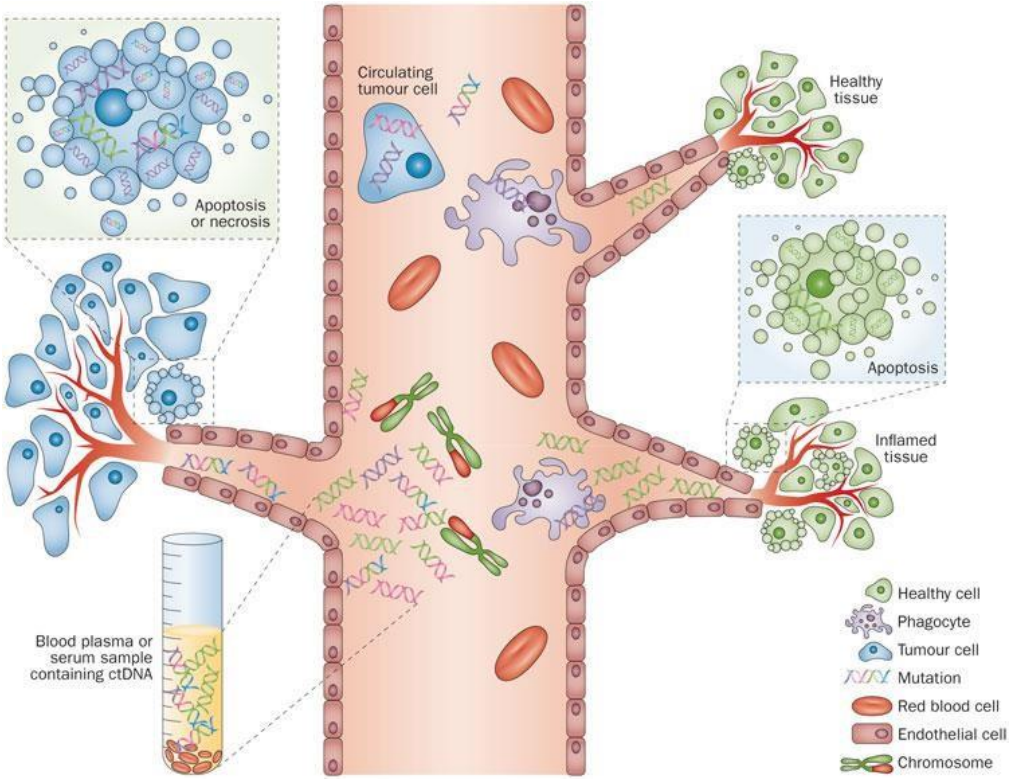
A)

B)



NGS:- Applications and data analysis: Liquid Biopsy

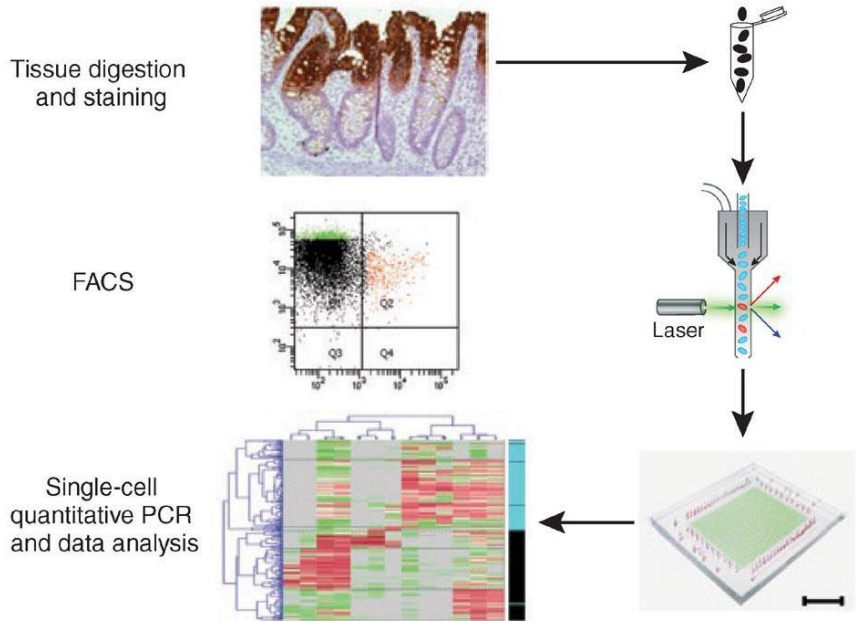


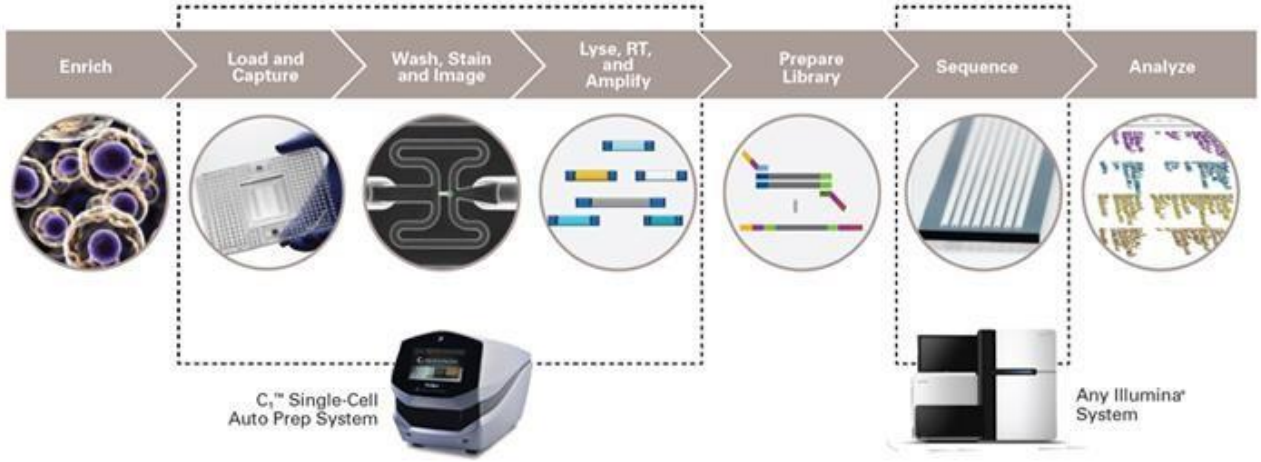




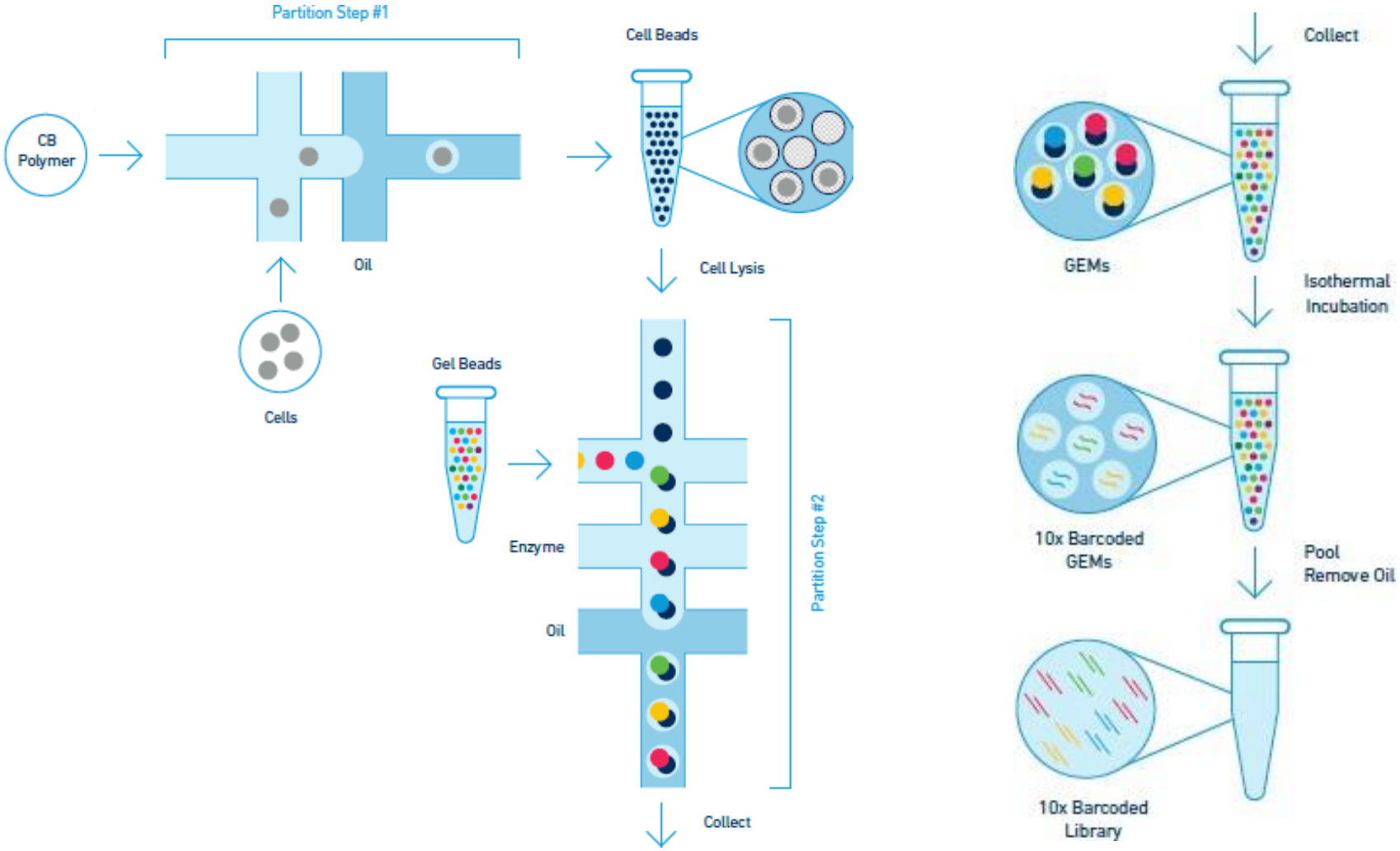
Applications:
Cancer detection
Prenatal anormalities
Drug response

NGS:- Applications and data analysis: Single Cell -omics



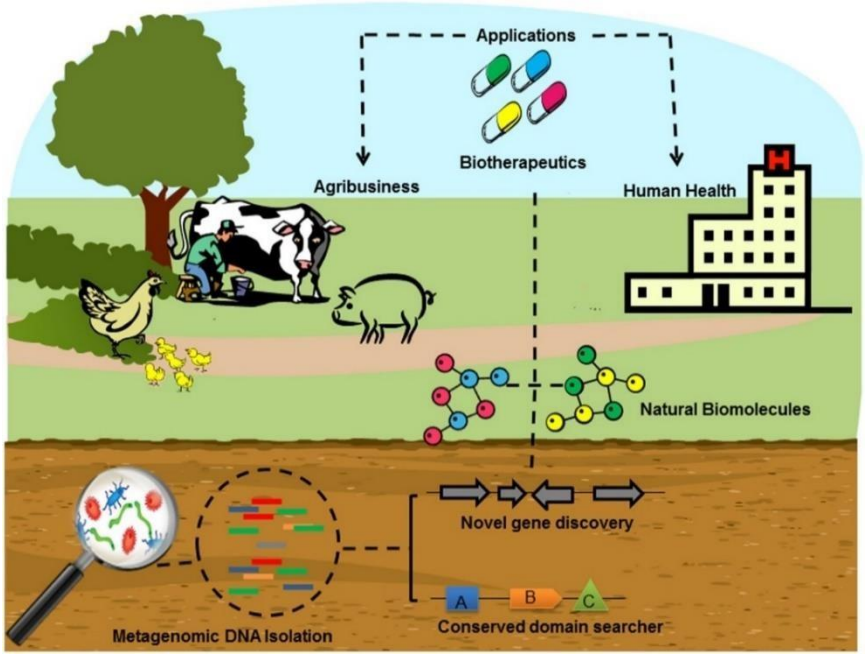


Single Cell -omics : 10x genomics

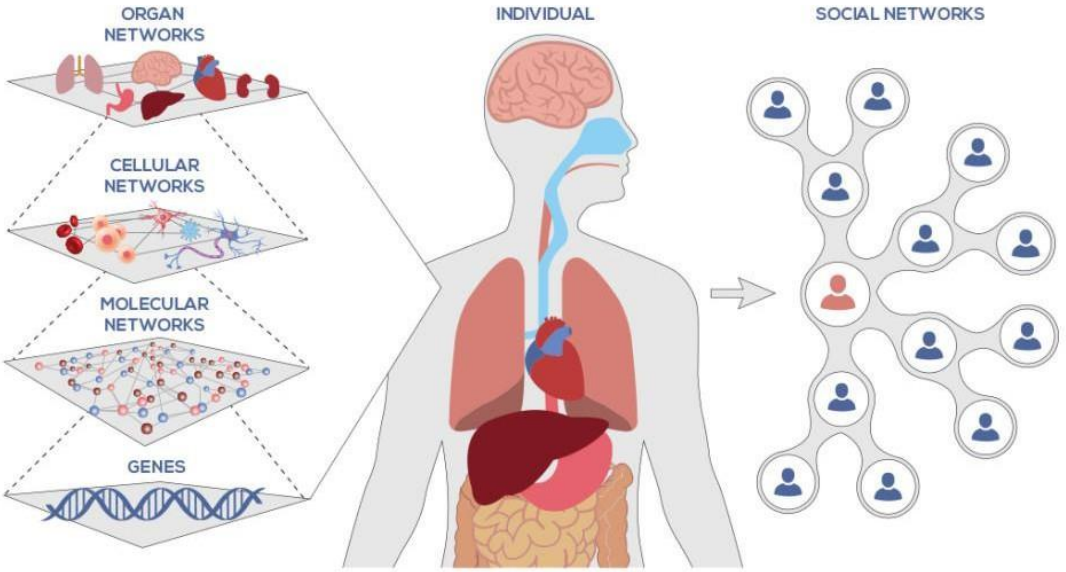


https://www.10xgenomics.com/solutions/single-cell/

Systems Biology

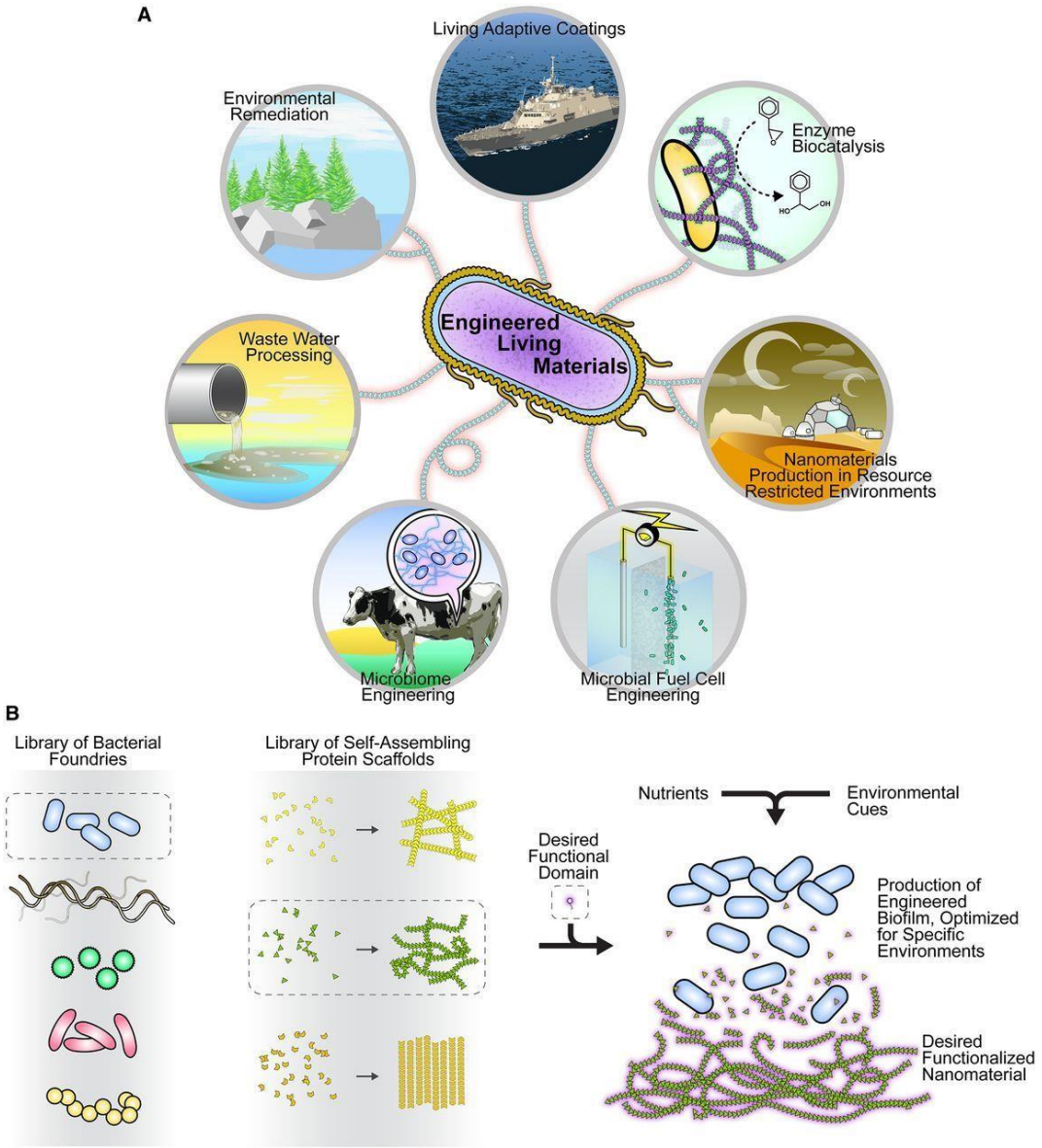


Systems biology is the computational and mathematical modeling of complex biological systems. It is a biology-based interdisciplinary field of study that focuses on complex interactions within biological systems

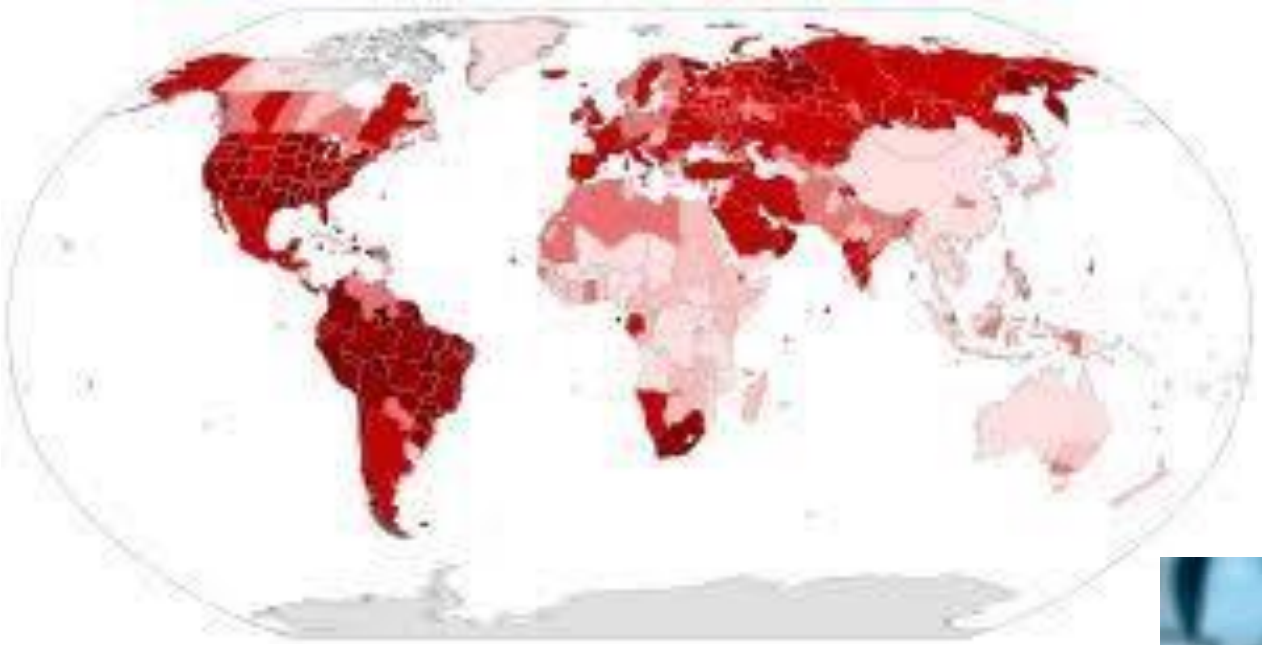


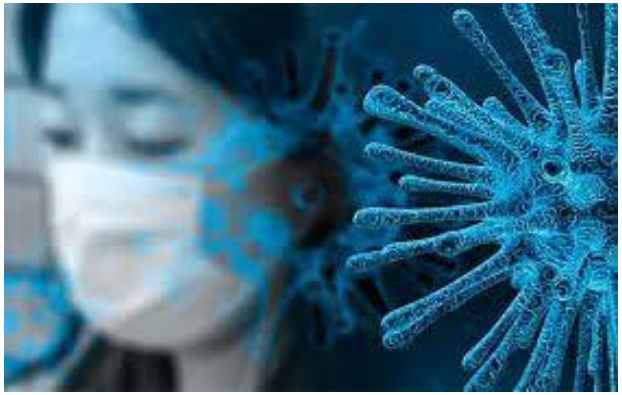
Synthetic Biology

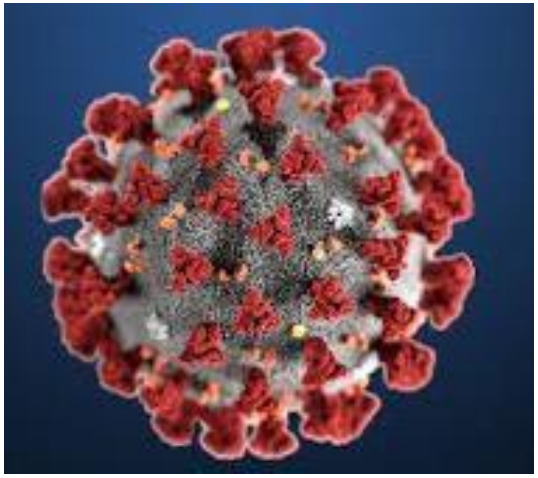
Synthetic biology is a new interdisciplinary area that involves the application of engineering principles to biology. It aims at the (re-)design and fabrication of biological components and systems that do not already exist in the natural world.



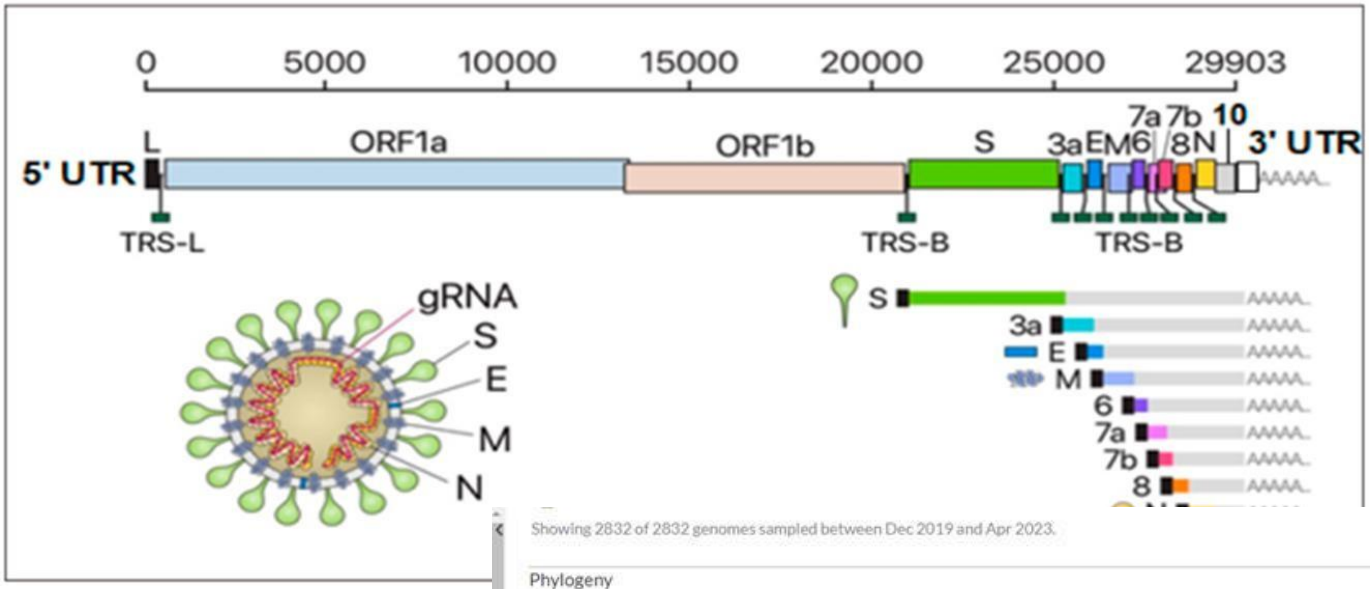
COVID 19

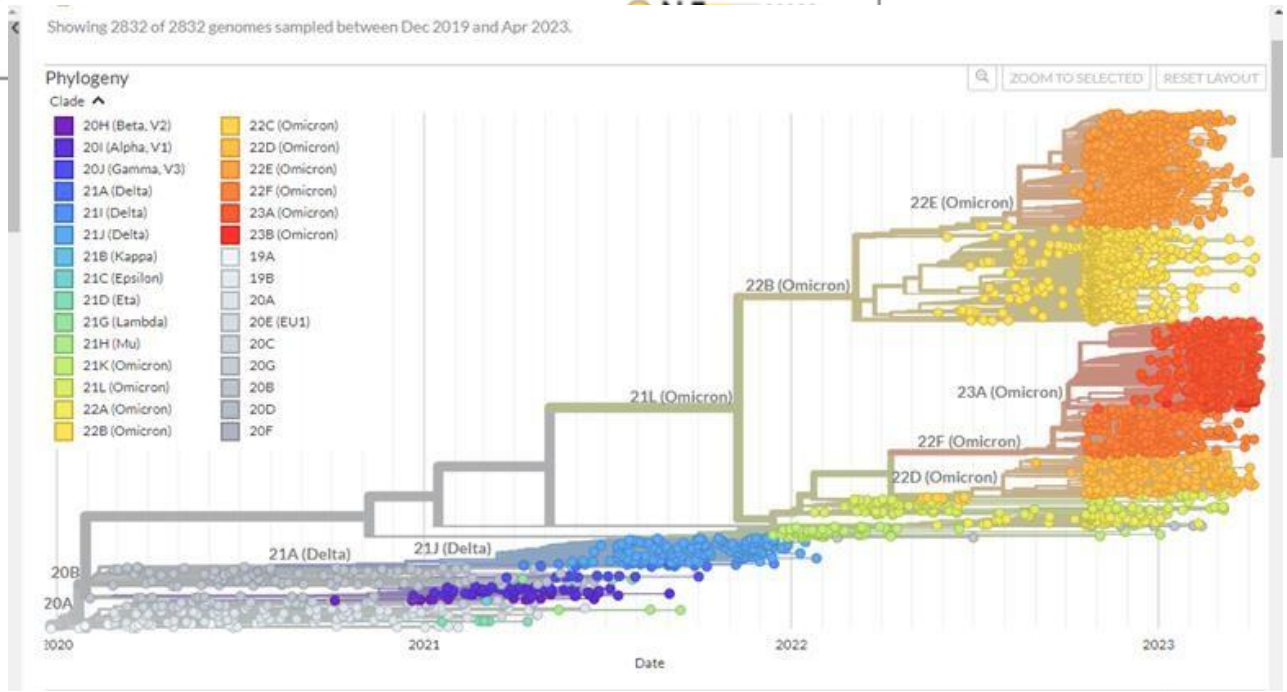






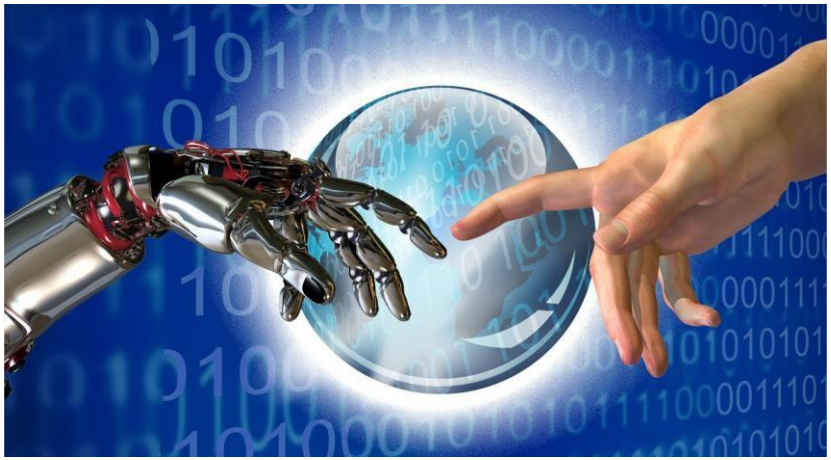
COVID 19





https://nextstrain.org/ncov/global

Science and technology





GRACIAS!!!!!!!

Preguntas??

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